Predict and Annotate Functions for Noncoding Variants

Lecture 5

Outline

- Why Studying Noncoding Variants
- DEEPSEA (Zhou J. & Troyanskaya O., Nature Methods, 2015)
- FAVOR (Zhou H. et al. Nucleic Acids Res, 2022)
- Integrate with GWAS Data

Part of the slides are adapted from the slides by Anthony Gitter, Mark Craven, Colin Dewey and Daifeng Wang@wisc

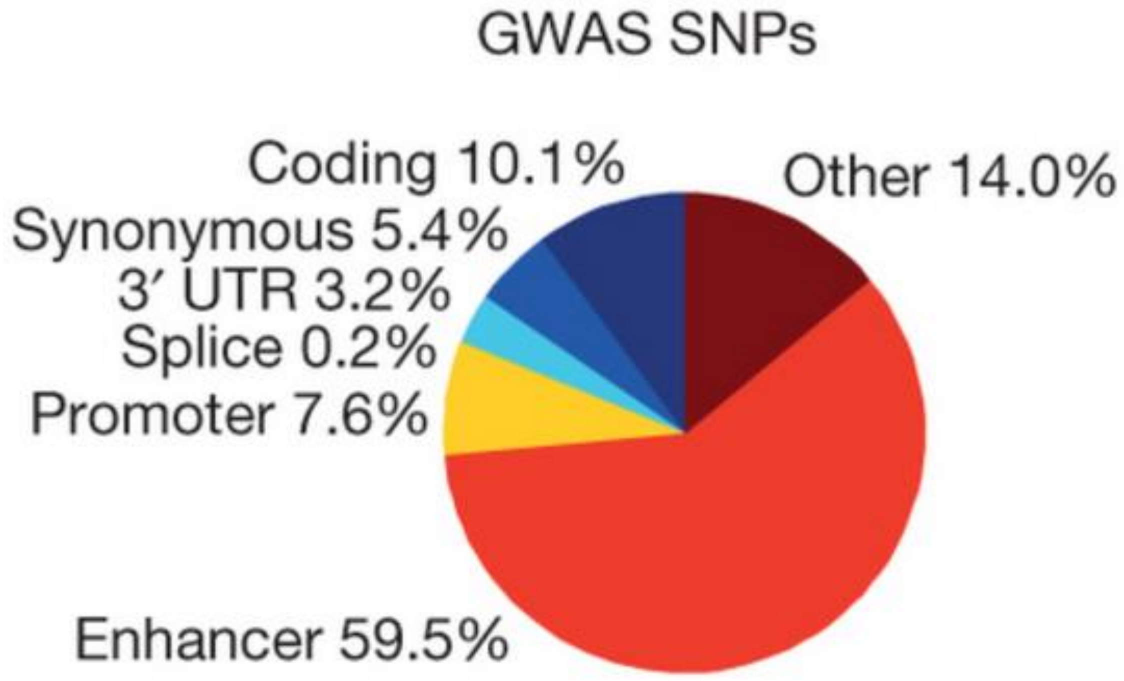
Why Studying Non-coding Variants?

- **Noncoding variants**: germline and somatic variants located in non-coding regions, which can play an important role in complex human diseases.
- Ways noncoding variants can be functional:
  - Disrupt DNA sequence motifs
    - Promoters, enhancers
  - Disrupt miRNA binding
  - Mutations in introns affect splicing
  - Indirect effects from the above changes
- Functional effects of non-coding variants can be studied by experiments and computational prediction.

Success of GWAS

As of 2023-05-20, the GWAS Catalog contains >6K publications with >519K top associations

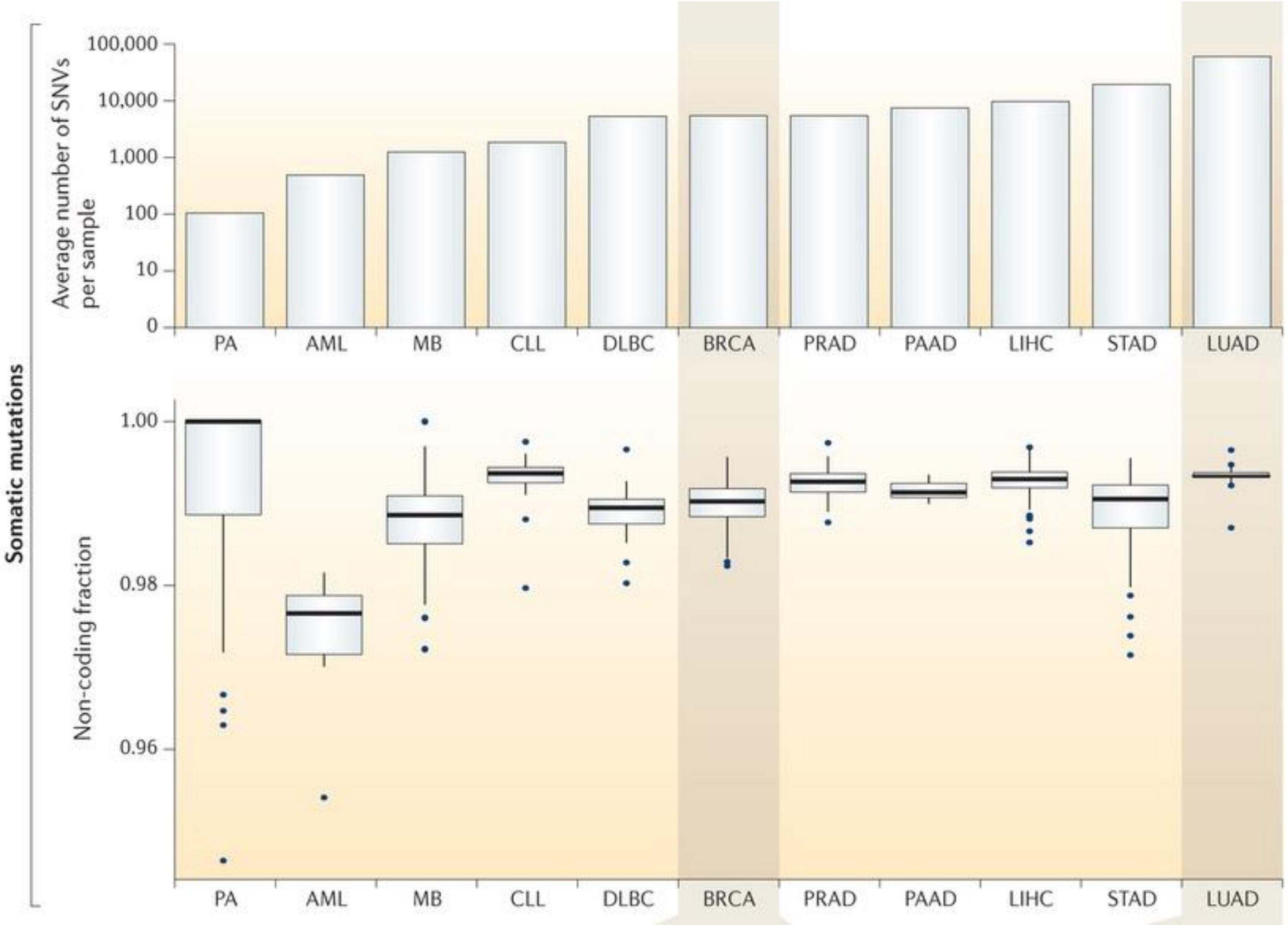
Annotations of GWAS Signals of 21 Autoimmune Diseases



Farh K.K. et al. Nature 2015

90% prioritized “causal” SNPs are noncoding

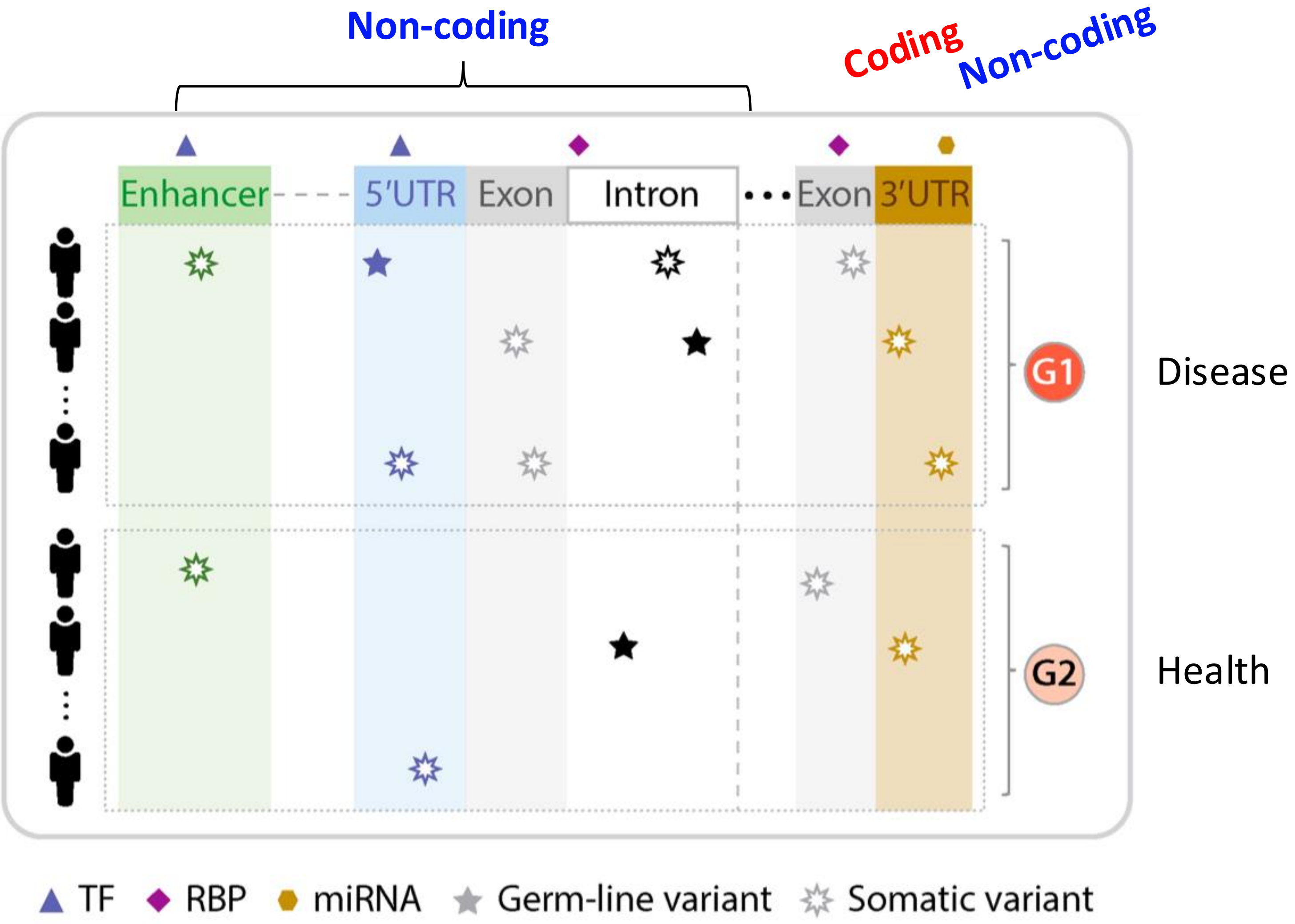
Almost all single nucleotide variants in cancer are noncoding



Khurana E. et al. *Nature Reviews Genetics* 2016

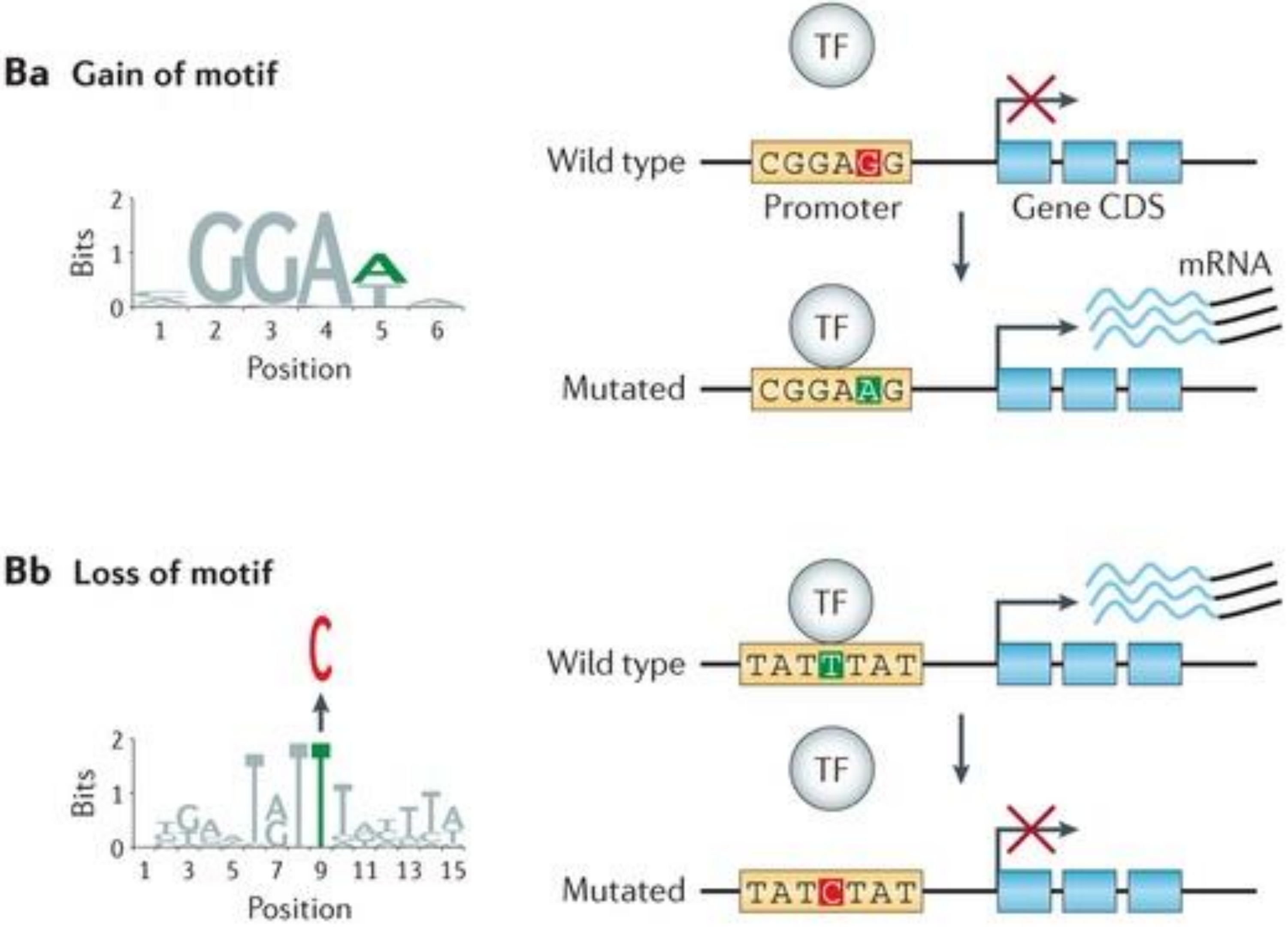
However, very few of these are driver mutations

How to link non-coding disease SNPs to genes?



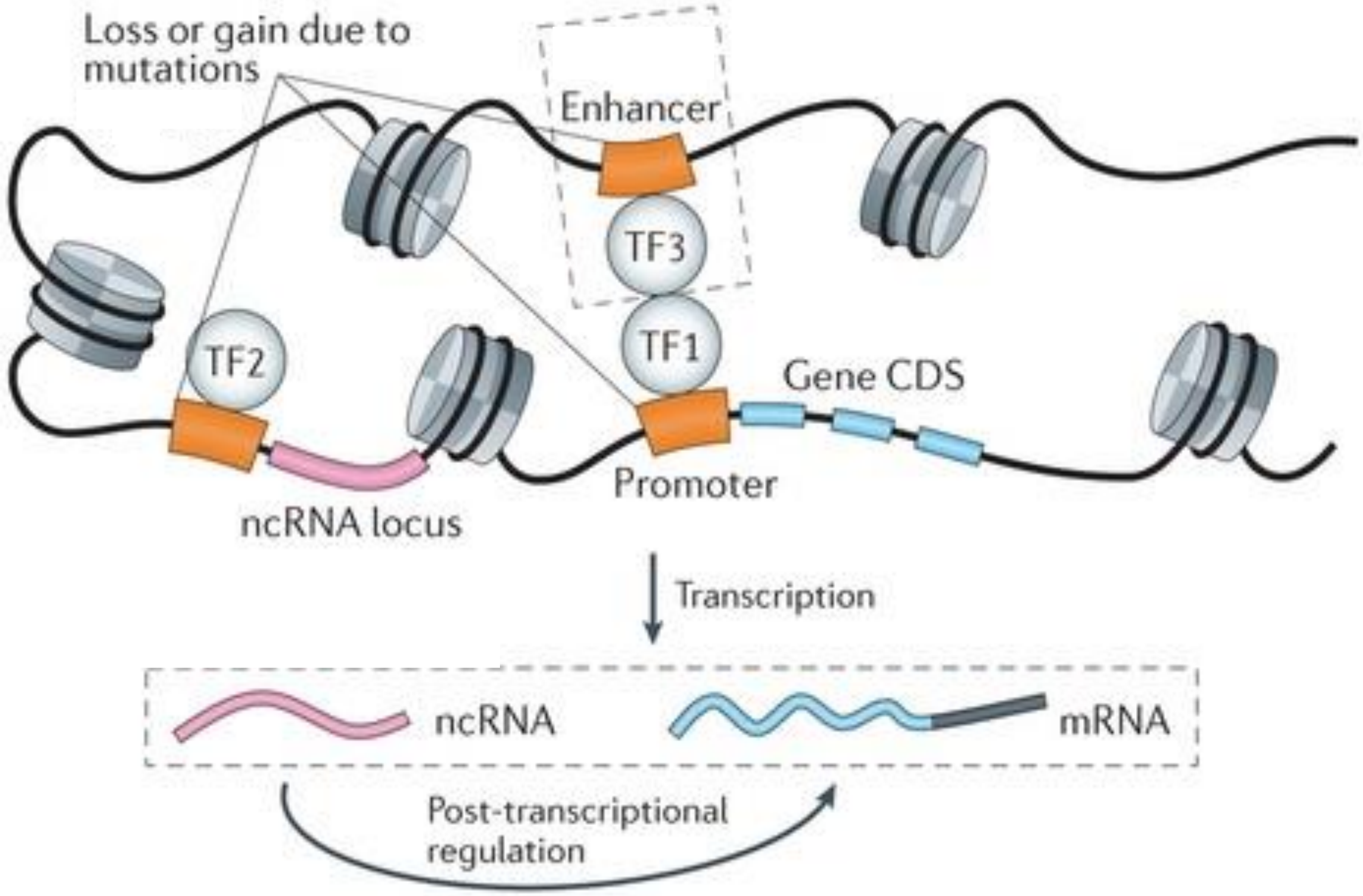
McGillivray P, et al. 2018.
Annu. Rev. Biomed. Data Sci. 1:153–80

Variants Altering Motifs



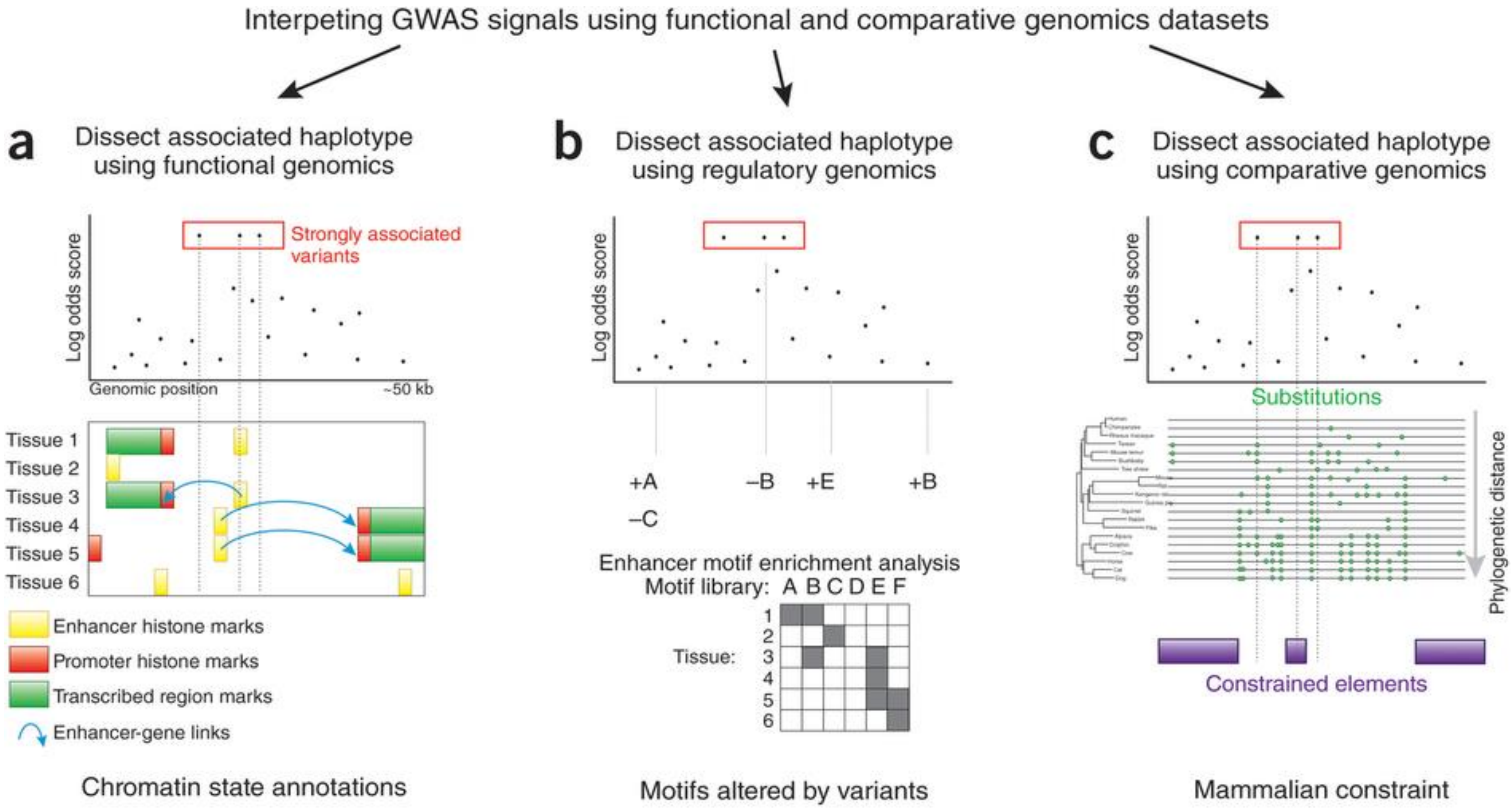
Khurana *Nature Reviews Genetics* 2016

Variants Affect Proximal and Distal Regulators



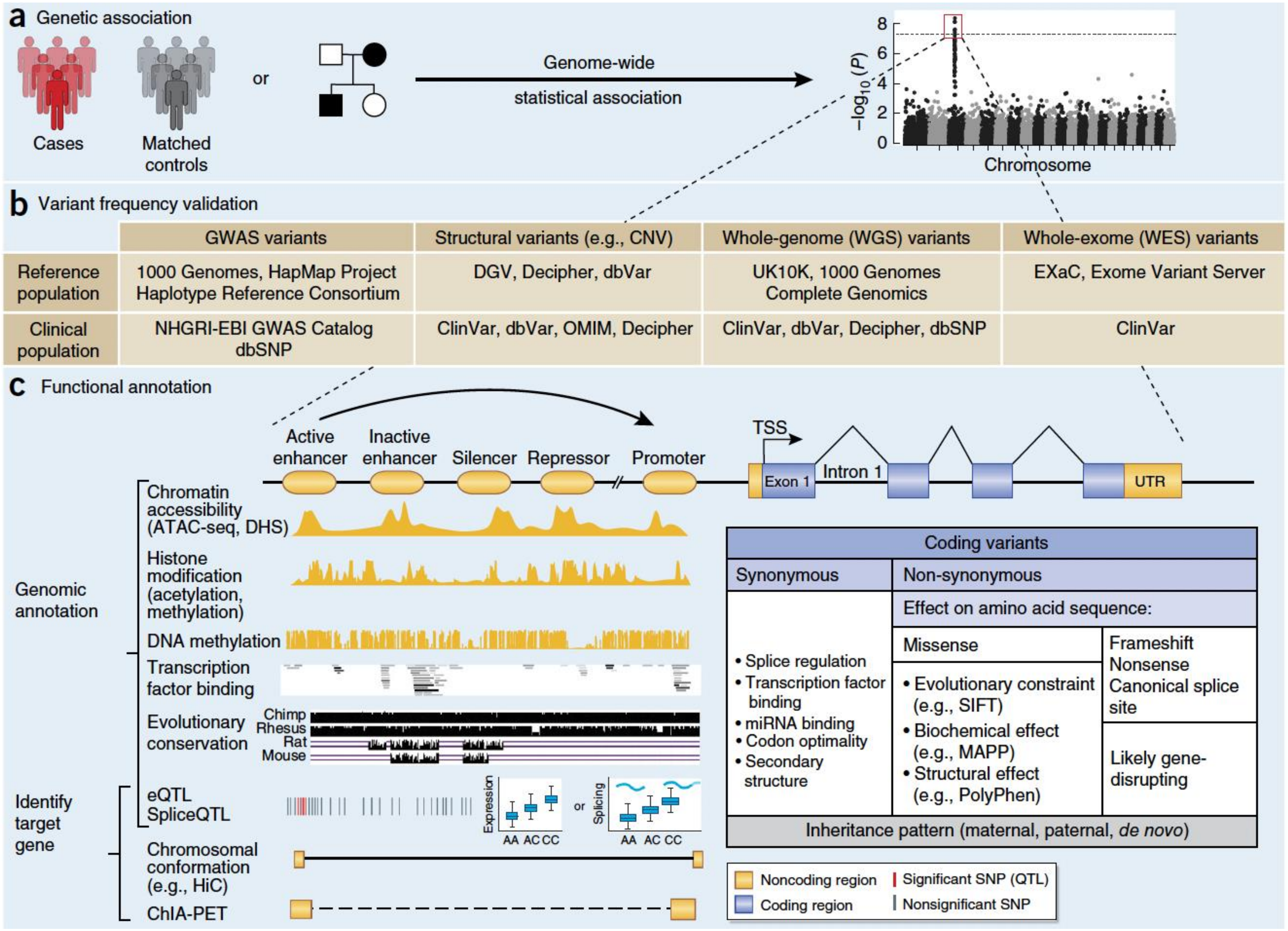
Khurana *Nature Reviews Genetics* 2016

Evidence Used to Prioritize Noncoding Variants



Ward and Kellis *Nature Biotechnology* 2012

Functional Genomics from Variants to Genes



Gandal et al., Nature Neuroscience, 2016

Disease-associated genomic variants

How do variants function?

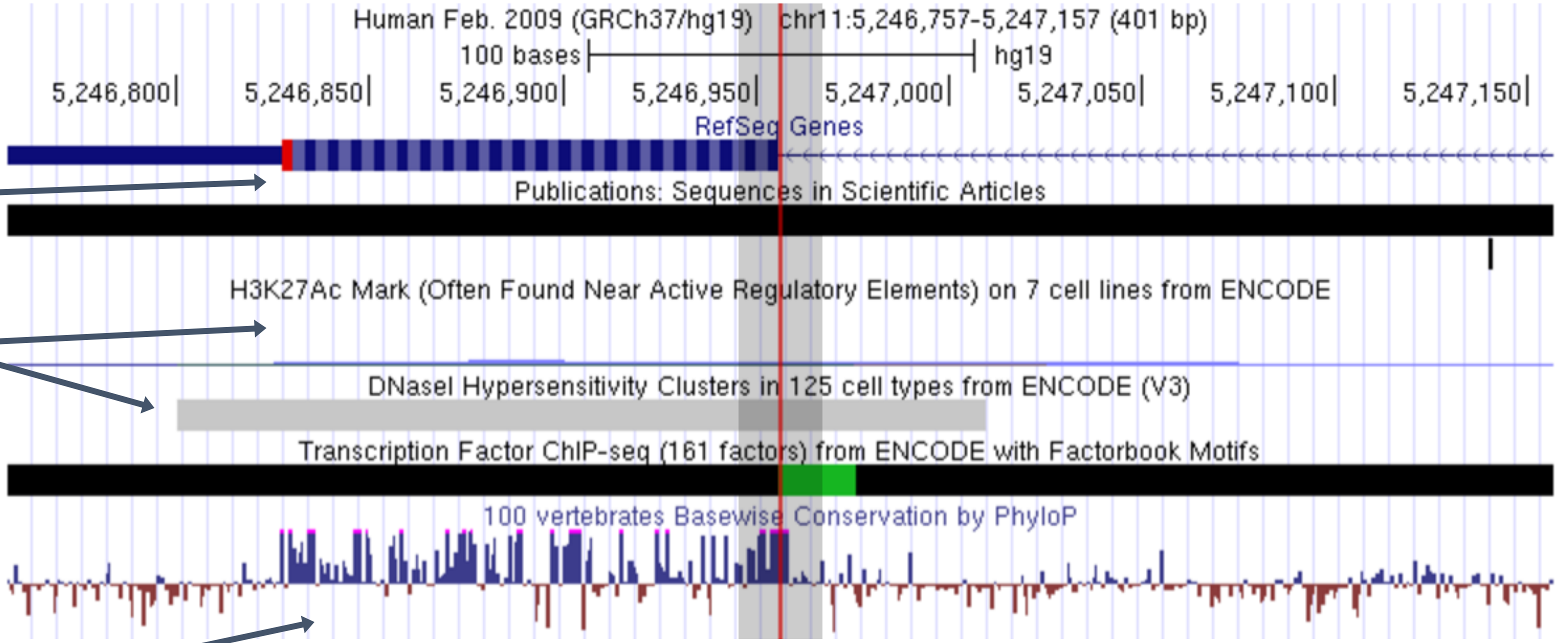
Visualizing Evidence

https://regulomedb.org/regulome-search/

Data supporting chr11:5246957 (rs33914668)

Summary of evidence → **Score: 2a**

Likely to affect binding

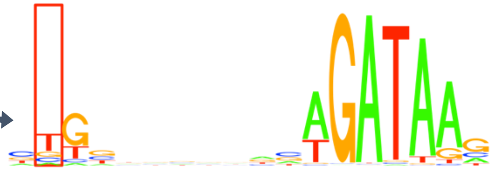


Genes

Epigenetic annotations

Conservation

Affected motifs



Boyle *Genome Research* 2012

Prioritizing GWAS Signals with Epigenetics Summary

- Disrupted regulatory elements are one of the best understood effects of noncoding SNPs
- Make use of extensive epigenetic datasets
- Similar strategies have actually worked
  - rs1421085 in *FTO* region for obesity
  - Claussnitzer *New England Journal of Medicine 2015*
- Epigenetic data at a genomic position is often in the presence of the reference allele
  - Don’t have measurements for the SNP allele
  - Pairing epigenetic data with genotype data

DeepSEA: Deep Learning-Based Sequence Analyzer

- Given:
  - A sequence variant and surrounding sequence context
  - Large-scale chromatin-profiling data (for model training)

- Do:
  - Predict TF binding, DNase hypersensitivity, and histone modifications in multiple cell and tissue types
  - Predict variant functionality with single-nucleotide sensitivity

Zhou and Troyanskaya *Nature Methods* 2015

DeepSEA: Deep Learning-Based Sequence Analyzer

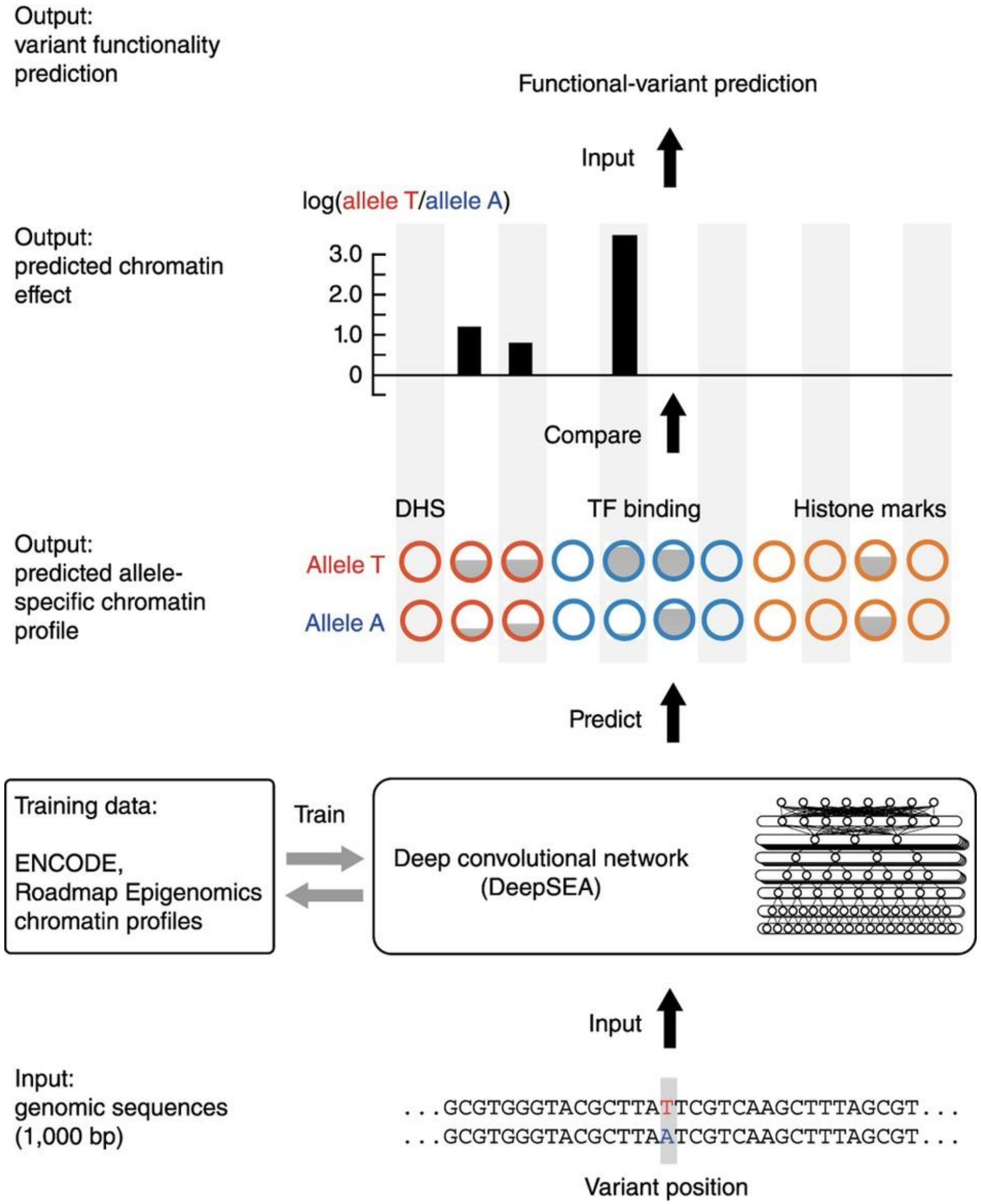
- First directly learn regulatory sequence code from genomic sequence by learning to simultaneously predict large-scale chromatin-profiling data, including TF binding, DNase I hypersensitivity, and histone-mark profiles
- Three major features
  - Integrating sequence information from a wide sequence context (e.g., 1kbp): *sequence surrounding the variant position determines the regulatory properties of the variant*
  - Learning sequence code at multiple spatial scales with a hierarchical architecture: *allows scaling to such long sequence input and learning sequence dependencies at multiple scales*
  - Multitask joint learning of diverse chromatin factors sharing predictive features: *computational efficient, allows predictive strength to be shared across a wide range of chromatin feature profiles for TF binding, DNase I hypersensitive sites (DHSs), and histone marks*

Zhou and Troyanskaya *Nature Methods* 2015

Schematic Overview of DeepSEA: Predicting chromatin effects of noncoding variants.

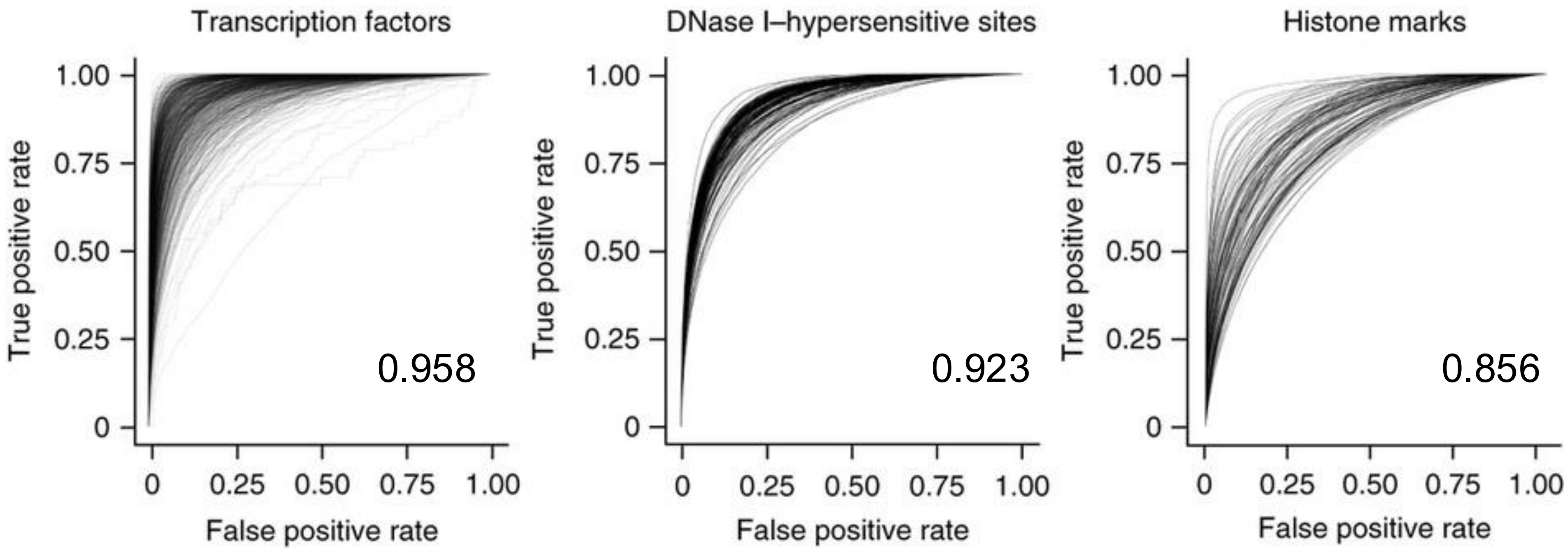
For model training: Genome-wide chromatin profiles from ENCODE and Roadmap Eigenomics projects, including 690 TF binding profiles for 160 different TFs, 125 DHS profiles and 104 histone-mark profiles.

In total, 521.6 Mbp of the genome (17%) were found to be bound by at least one measured TF and were used as a regulatory information–rich and challenging set for training our DeepSEA regulatory code model.



Predicting Epigenetic Annotations

- Compute median AUROC for three types of classes



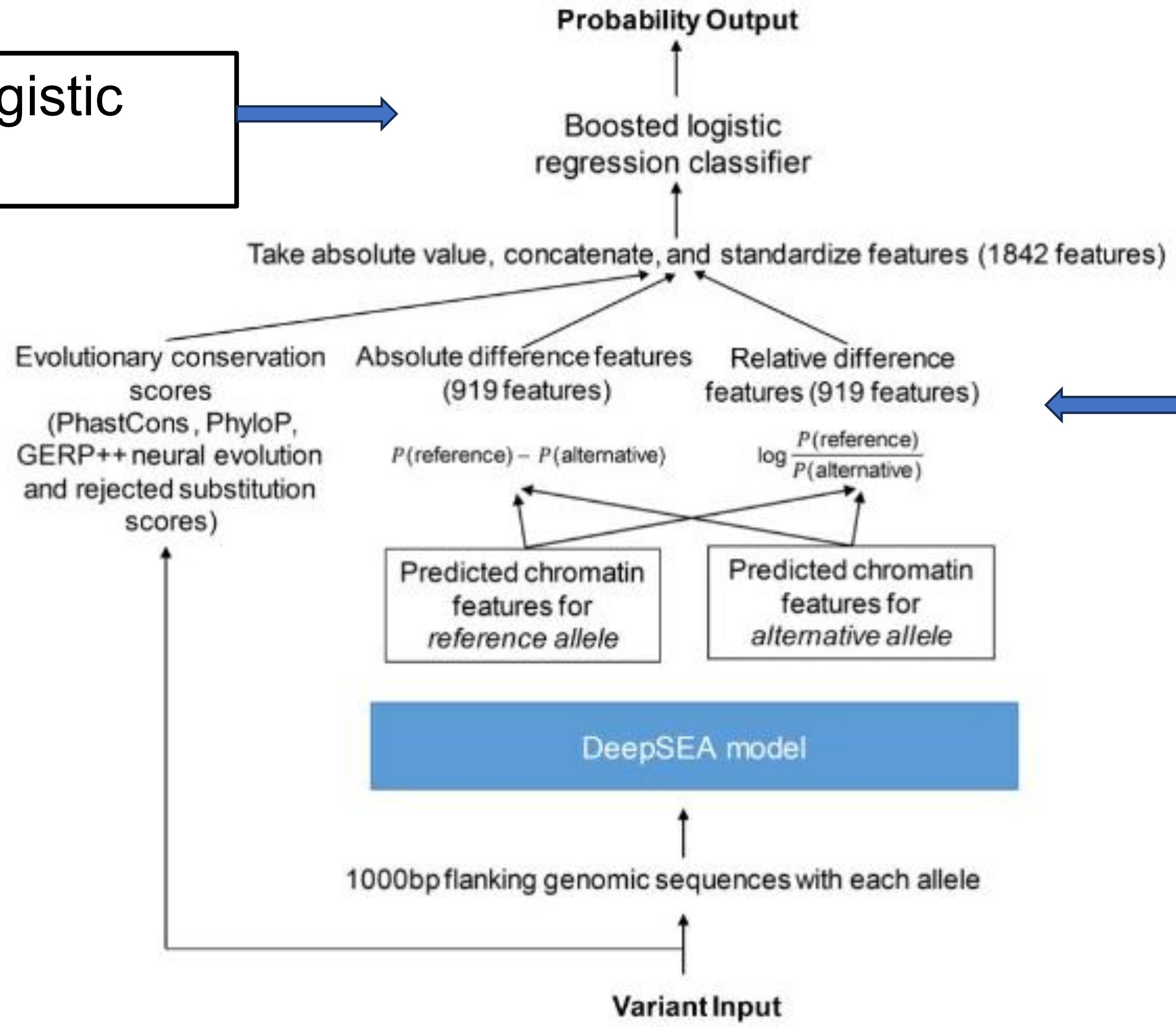
Zhou and Troyanskaya *Nature Methods* 2015

Predicting Functional Variants

- Can predict epigenetic signal for any novel variant (SNP, insertion, deletion)
- Define novel features to classify variant functionality
  - Difference in probability of signal for reference and alternative allele
- Train on SNPs annotated as regulatory variants in GWAS and eQTL databases
  - Human Gene Mutation Database (**HGMD**) annotated noncoding regulatory mutations
  - Noncoding eQTLs from the **GRASP** (Genome-Wide Repository of Associations between SNPs and Phenotypes) database
  - Noncoding trait-associated SNPs identified in GWAS studies from the US National Human Genome Research Institute's **GWAS Catalog**
  - Negative 'nonfunctional' variant standards: 1000 Genomes Project SNPs with controlled minor allele frequency distribution in 1000 Genomes population

Predicting Functional Variants: Prioritize Functional SNPs from GWAS

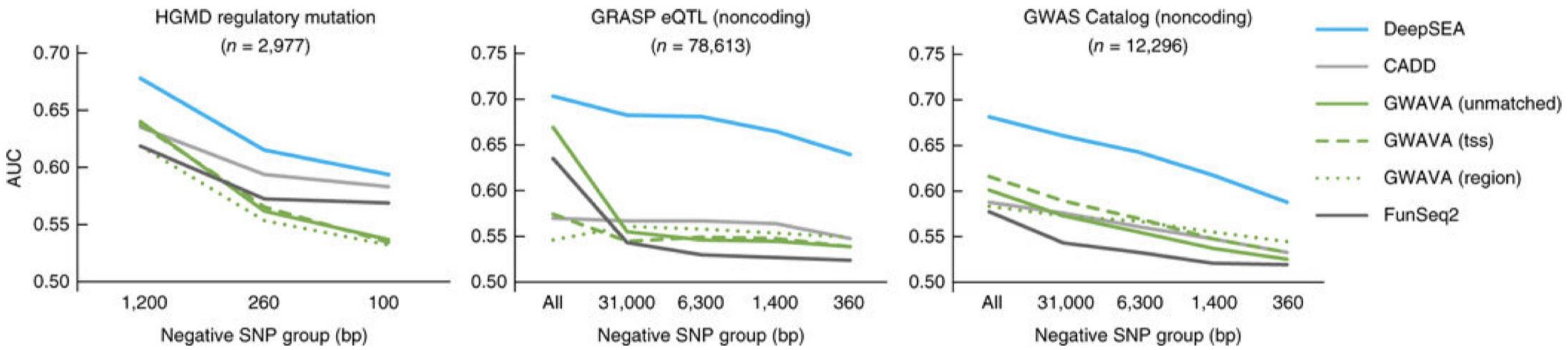
Boosted logistic regression



Conservation and predicted epigenetic impact of variant as features

Zhou and Troyanskaya *Nature Methods* 2015

Fig. 3. Sequence-based prioritization of functional noncoding variants.



The x axes show the average distances of negative-variant groups to a nearest positive variant. The “All” negative-variant groups are randomly selected negative 1000 Genomes SNPs.

DeepSEA Summary

- <u>Advantages</u>
  - Ability to predict how unseen variants affect regulatory elements
  - Accounts for sequence context of motif
  - Parameter sharing with convolutional layers
  - Multitask learning to improve hidden layer representations

- <u>Limitations</u>
  - Does not extend to new types of cells and tissues
  - AUROC is misleading for evaluating genome-wide epigenetic predictions

Predicting New TF-cell Type Pairs

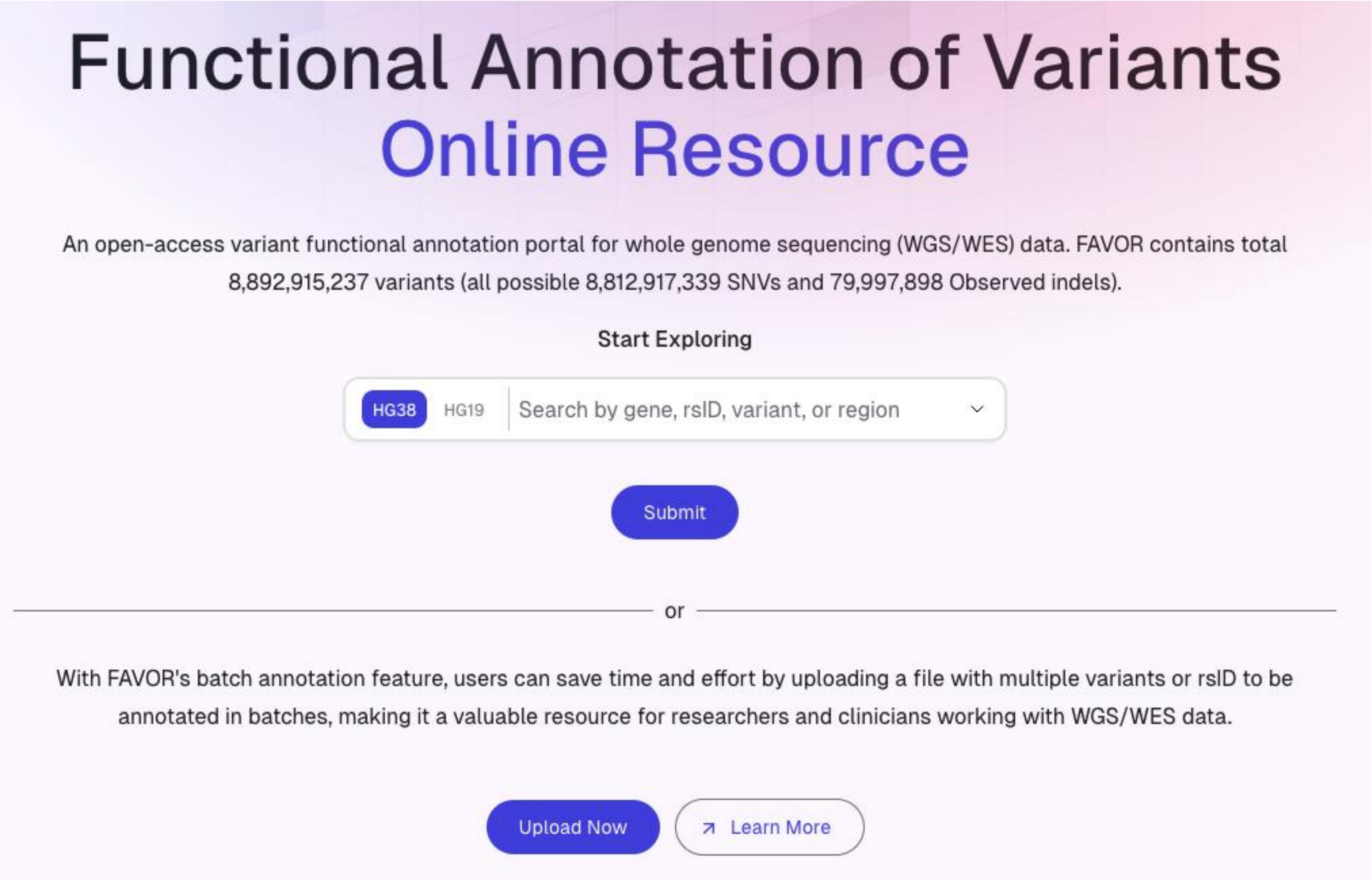
Training data

TF	Cell type
A	1
A	2
A	3
B	3
B	4
C	1
C	4

- DeepSEA cannot predict pairs not present in training data
  - Can predict TF A in cell type 1
  - Not TF A in cell type 4
- New methods can
  - TFImpute
  - FactorNet
  - Virtual ChIP-seq

FAVOR Annotator (https://favor.genohub.org)

Zhou H. et. al. *Nucleic Acids Res 2022 Nov 9; gkac966. DOI: 10.1093/nar/gkac966.*



Example Search of SNP rs7412 on CHR19/APOE

Chat with FAVOR-GPT

FAVOR-GPT | GPT 4o Mini

tell me function about rs7412

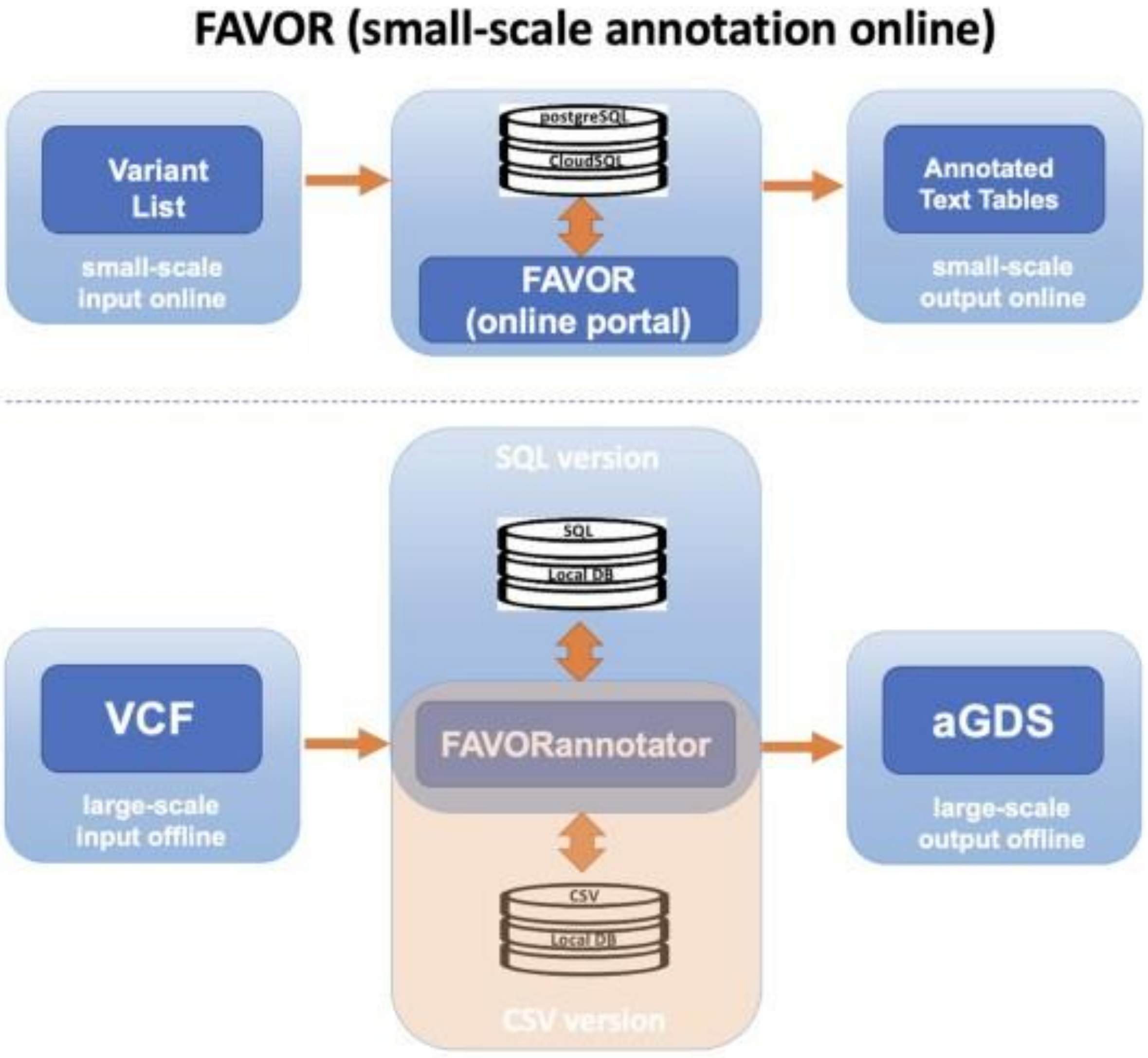
Successfully fetched additional information:

This data has been incorporated into the AI's knowledge.

The variant rs7412 is located on chromosome 19 at position 44908822. It is a nonsynonymous single nucleotide variant (SNV) that results in an amino acid change from Arginine (R) to Cysteine (C) at position 176 (p.R176C) of the APOE gene.

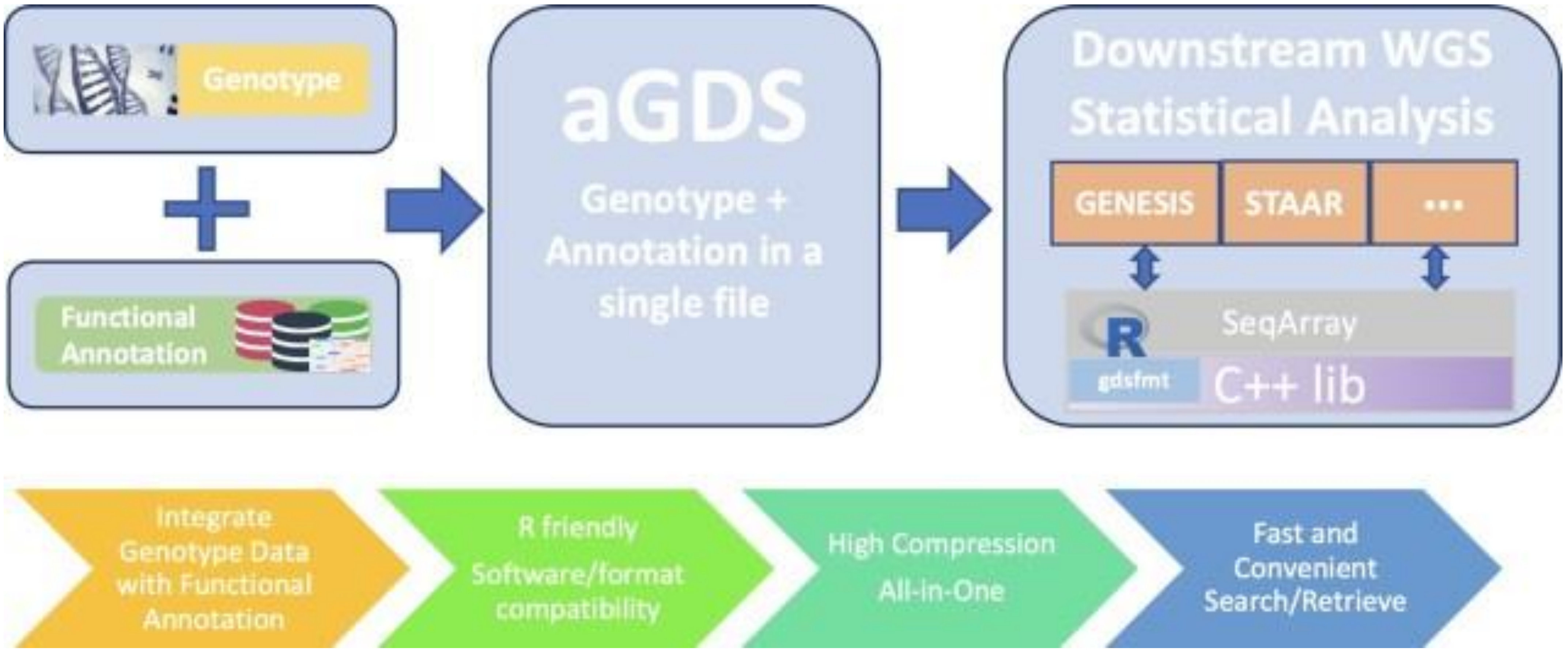
1. **Gene**: APOE
2. **Variant Type**: Exonic, nonsynonymous SNV
3. **Clinical Significance**: Associated with drug response and classified as pathogenic for conditions like Hypercholesterolemia and Warfarin response.
4. **Allele Frequency**: Approximately 7.81% in the general population.
5. **Polyphen2 Score**: 1 (probably damaging)
6. **SIFT Score**: 0 (deleterious)
7. **CADD Phred Score**: 25.3 (highly deleterious)

Features of FAVOR batch annotation and FAVORannotator.



FAVORannotator (large-scale annotation):
R-script to annotate any WGS/WES study using the FAVOR database

Features of the annotated Genomic Data Structure (aGDS) format.



How to integrate such predicted functional annotations of non-coding variants with GWAS data?

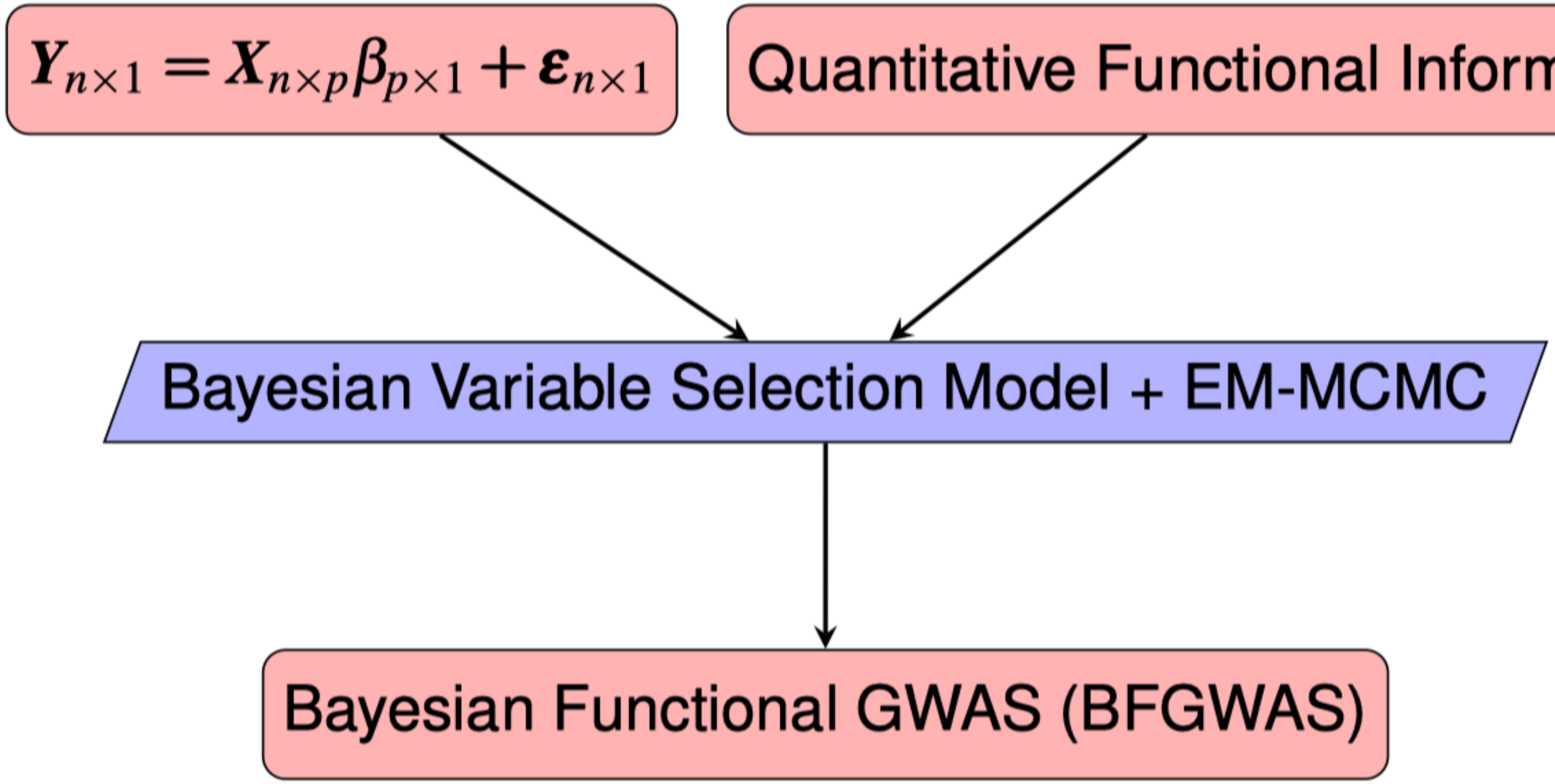
- Interpretate GWAS results?
- Link GWAS signals to function genes?
- Prioritize potential “causal” variants?

BFGWAS_QUANT: Bayesian Functional GWAS Method Accounting for Multivariate Quantitative Functional Annotations

Motivations

- Account for linkage disequilibrium (LD, correlation among genetic variants) to fine-map independent association
- Account for multivariate quantitative functional annotations to prioritize potential causal variants
- Use publicly available summary-level GWAS data of large sample sizes
- Understand biological mechanisms underlying genetic associations

BFGWAS Diagram



J. Yang et. al. AJHG 2017.

Bayesian Hierarchical Model Framework

Multivariable linear regression model with **Standardized** phenotype and genotype vectors:

$$\boldsymbol{Y}_{n\times 1} = \boldsymbol{X}_{n\times p}\beta_{p\times 1} + \boldsymbol{\varepsilon}_{n\times 1}, \quad \varepsilon \sim MN(0, I). \qquad (1)$$

Prior:

- $\beta_i \sim \pi_i N(0, \frac{1}{n}\tau_\beta^{-1}) + (1-\pi_i)\delta_0(\beta_i);\ i = 1, \cdots, p$
- With augmented quantitative functional annotation data vector $\boldsymbol{A_i} = (1, A_{i,1}, \cdots, A_{i,J})$ for variant $i = 1, \cdots, p$,

$$logit(\pi_i) = \boldsymbol{A}_i'\boldsymbol{\alpha};\ \pi_i = \frac{e^{A_i'\boldsymbol{\alpha}}}{1+e^{A_i'\boldsymbol{\alpha}}};\ \boldsymbol{\alpha} = (\alpha_0, \alpha_1, \cdots, \alpha_J)$$

- Introduce a latent indicator vector $\boldsymbol{\gamma}_{p\times 1}$, equivalently

$$\gamma_i \sim Bernoulli(\pi_i),\ \boldsymbol{\beta}_{-\boldsymbol{\gamma}} \sim \delta_0(\cdot),\ \boldsymbol{\beta}_{\boldsymbol{\gamma}} \sim MVN_{|\boldsymbol{\gamma}|}(0, \frac{1}{n}\tau_\beta^{-1}\boldsymbol{I}_{\boldsymbol{\gamma}})$$

Shared Parameters by Genome-wide Variants

- Fix $\tau_\beta \in (0, 1]$: Inverse of effect size variance in the multivariable regression model.
  - $\tau_\beta = 1$ assumes same prior effect size variance as the marginal effect sizes.
  - $\tau_\beta < 1$ assumes larger magnitude for effect sizes in the multivariable model than the marginal ones.

- $\alpha_j \sim N(0, 1), j = 1, \cdots, J$: Enrichment parameters

- Fix $\alpha_0 \in (-13.8, -9)$ to induce a sparse model.
  - $\alpha_0 = -13.8$ assumes prior causal probability 10^{-6} when $\alpha_j = 0, j = 1, \cdots, J$.

Parameters of Interest

- Enrichment parameters:
  - $(\alpha_1, \cdots, \alpha_J)$: for J annotations
- Variant-specific parameters (association evidence):
  - β_i: Effect-size
  - $\widehat{\pi}_i = E[\gamma_i]$: Causal Posterior Probability (CPP)
- Region-level (Association evidence):
  - `Regional_CPP` = $E[max(\gamma_{i_1}, \cdots, \gamma_{i_k})]$: Regional probability of having at least one causal variant
  - `Sum_CPP` = $\sum_{i=1}^{p} \widehat{\pi}_i I(\pi_i > 0.01)$: Expected number of causal SNPs

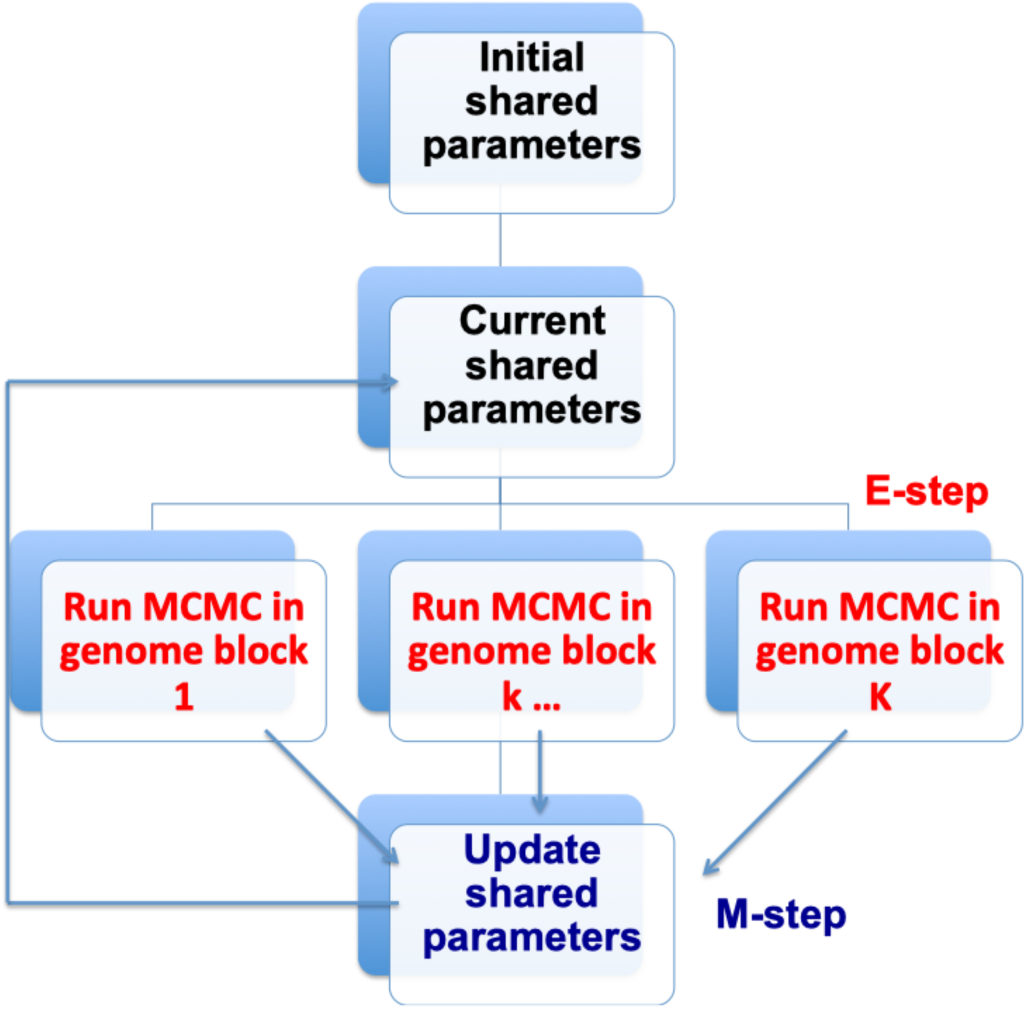
Bayesian Inference

- Joint posterior distribution

$$P(\boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\pi}(\boldsymbol{\alpha}) | \boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{A}) \propto \tag{2}$$
$$P(\boldsymbol{Y}|\boldsymbol{X}, \boldsymbol{\beta}, \boldsymbol{\gamma}) P(\boldsymbol{\beta}|\boldsymbol{\gamma}, \tau_\beta) P(\boldsymbol{\gamma}|\boldsymbol{\pi}(\boldsymbol{\alpha}), \boldsymbol{A}) P(\boldsymbol{\pi}(\boldsymbol{\alpha}))$$

- Product of Likelihood and Prior
- MCMC by Metropolis-Hastings algorithm with a proposal strategy for $\boldsymbol{\gamma}$
- Challenges of Standard MCMC: Memory Usage and Convergence Rate for $\sim 10M$ genome-wide variants

EM-MCMC Algorithm



Enable Genome-wide Analysis

Improve MCMC Convergence Rate

MCMC Algorithm

Given shared parameters $(\alpha_0, \alpha_1, \cdots, \alpha_J), \tau_\beta$:

- Propose a new indicator vector γ
- Posterior conditional distribution for $\boldsymbol{\beta}_{|\gamma|}$:

$$P(\boldsymbol{\beta}_{|\gamma|} | \boldsymbol{Y}, \boldsymbol{X}, \boldsymbol{\gamma}, \tau_\beta) \sim MVN_{|\boldsymbol{\gamma}|}(\boldsymbol{\mu}_{\boldsymbol{\beta}_{|\gamma|}}, \boldsymbol{\Sigma}_{\beta_{|\gamma|}});$$

$$\boldsymbol{\mu}_{\boldsymbol{\beta}_{|\gamma|}} = \boldsymbol{\Sigma}_{\beta_{|\gamma|}} \boldsymbol{X}^T \boldsymbol{Y},\ \boldsymbol{\Sigma}_{\beta_{|\gamma|}} = \frac{1}{\boldsymbol{n}} (\boldsymbol{R} + \tau_\beta \boldsymbol{I}_{m \times m})^{-1},\ \boldsymbol{R} = \frac{1}{\boldsymbol{n}} \boldsymbol{X}^T \boldsymbol{X}$$

- Conditional posterior likelihood:

$$P(\gamma | Y, X, \pi(\alpha), \tau_\beta) \propto$$

$$\sqrt{|\boldsymbol{\Sigma}_{\beta_{|\gamma|}}|} \cdot (n\tau_\beta)^{\frac{m}{2}} \cdot \exp\left\{ -\frac{n}{2} + \frac{1}{2} (\boldsymbol{X}^T \boldsymbol{Y})^T \boldsymbol{\Sigma}_{\beta_{|\gamma|}} (\boldsymbol{X}^T \boldsymbol{Y}) \right\} \cdot \prod_{i=1}^{P} P(\gamma_i | \pi(\alpha_i))$$

Update Enrichment Parameters by Maximum A Posteriori (MAP)

- The expected log-posterior-likelihood function of $\boldsymbol{\alpha}$:

$$l(\boldsymbol{\alpha}) = E_{\gamma}[ln(P(\boldsymbol{\alpha}|\boldsymbol{\gamma},A))] \propto \sum_{i=1}^{p}\left[\hat{\gamma}_i ln\left(e^{A_i'\boldsymbol{\alpha}}\right) - ln\left(1+e^{A_i'\boldsymbol{\alpha}}\right)\right] - \frac{\boldsymbol{\alpha}'\boldsymbol{\alpha}}{2}$$

- Enrichment parameters $\boldsymbol{\alpha}$ are estimated by using the following gradient and hessian functions:

$$\frac{dl(\boldsymbol{\alpha})}{d\boldsymbol{\alpha}} = \sum_{i=1}^{p}\left[\widehat{\gamma}_i \boldsymbol{A}_i' - \left(1+e^{-A_i'\boldsymbol{\alpha}}\right)^{-1}\boldsymbol{A}_i'\right] - \boldsymbol{\alpha}'$$

$$\frac{d^2 l(\boldsymbol{\alpha})}{d\boldsymbol{\alpha}d\boldsymbol{\alpha}'} = -\sum_{i=1}^{p}\left[\frac{e^{-A_i'\boldsymbol{\alpha}}}{(1+e^{-A_i'\boldsymbol{\alpha}})^2}(\boldsymbol{A}_i\boldsymbol{A}_i')\right] - \boldsymbol{I}$$

Parameters of Interest

- Significant causal SNPs with $CPP > 0.1068$ (equivalent to p-value < 5×10^{-8}), effect size estimates $\widehat{\beta}_i$ and posterior causal probability $\widehat{\pi}_i$
- Estimates of enrichment parameters $(\alpha_1, \cdots, \alpha_J)$
- Sum of CPP estimates for variants with $\widehat{\pi}_i > 0.01$ estimates the number of expected GWAS signals

eQTL based Annotations

Derived from frontal cortex brain tissues and Microglia cells:

- Allcis-eQTL: Binary annotation denoting if a SNP is a significant cis-eQTL
- 95%CredibleSet: Binary annotation denoting if a SNP is in within a fine-mapped 95% credible set of cis-eQTL by CAVIAR
- MaxCPP: Maximum cis-CPP per SNP across all genes
- BGW_MaxCPP: Maximum CPP (cis- or trans-) per SNP across all genes derived by our BGW-TWAS method
- Microglia-eQTL : Binary annotation denoting if a SNP is a significant cis-eQTL of Microglia cell type

Histone Modifications based Annotations

Derived from the epigenomics data in the brain mid frontal gyrus region from the ROADMAP Epigenomics database:

- H3K4me1 (primed enhancers)
- H3K4me3 (promoters)
- H3K36me3 (gene bodies)
- H3K27me3 (polycomb regression)
- H3K9me3 (heterochromatin)

All binary annotations denoting if the SNP is located in the peak regions of the above histone modifications.

Application Studies by BFGWAS_QUANT

Phenotype: Clinical diagnosis of AD dementia

Individual-level GWAS Data

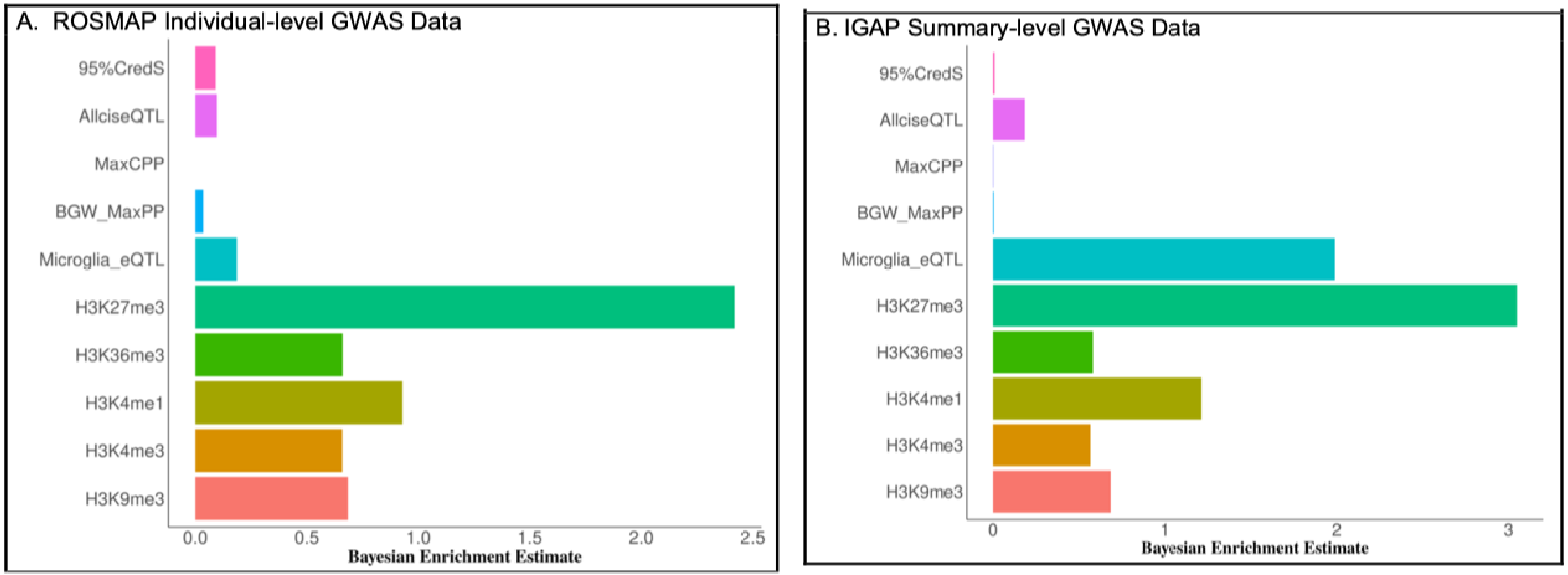
- Religious Orders Study and Rush Memory and Aging Project (ROS/MAP): 1,417 WGS samples of European ancestries
- Adjust for covariates: age, sex, smoking status, study index (ROS or MAP), and top 3 genotype PCs

IGAP Summary-level GWAS Data

- Meta-analysis summary data from 4 cohorts with ~ 54K samples.
- Reference LD was derived from the ROS/MAP WGS genotype data

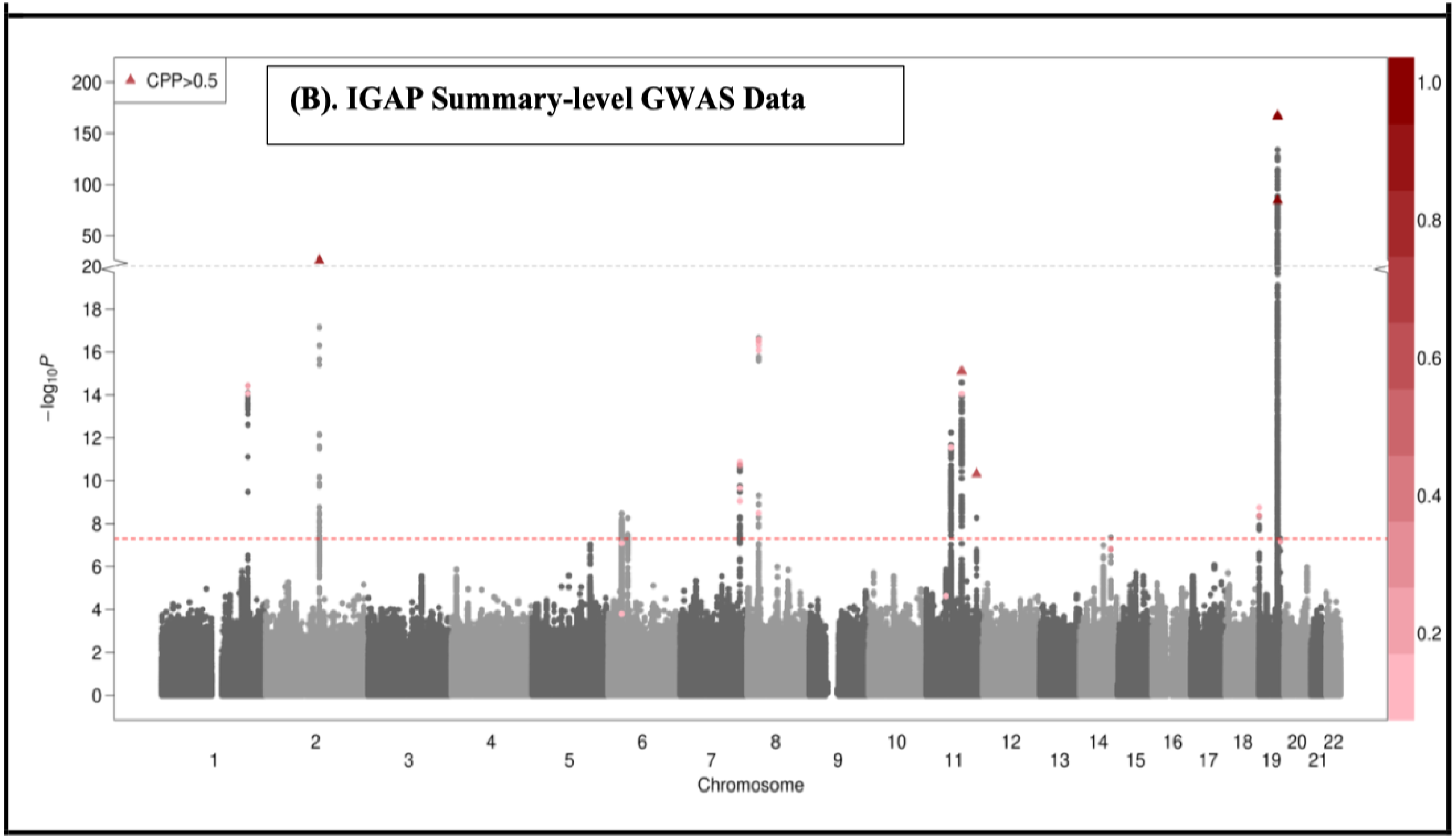
Considered 10 eQTL and histone modification based annotations

Estimates of Enrichment Parameters in Real Data



- **Microglia is a known related cell type in the brain for Alzheimer's disease.**
- **H3K27me3 is an epigenetic modification to the DNA packaging protein Histone H3, the tri-methylation of lysine 27 on histone H3 protein, which is associated with the downregulation of nearby genes via the formation of heterochromatic regions.**

BFGWAS Results of AD Dementia



Sum_CPP=54.3

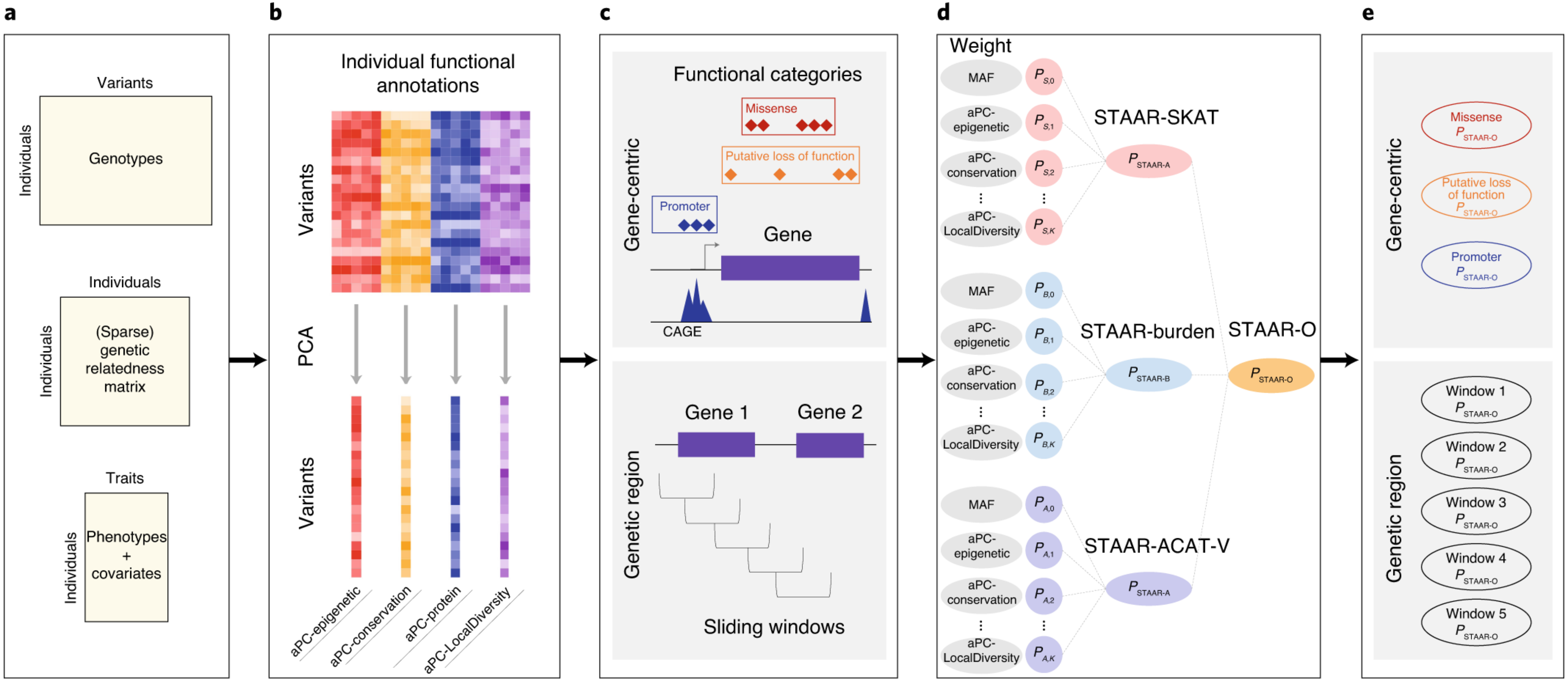
Table 2. Significant SNPs with Bayesian CPP > 0.1068 by BFGWAS_QUANT for studying AD using the IGAP summary-level GWAS data. SNPs with single variant test P-value $> 5 \times 10^{-8}$ were shaded in gray.

CHR	rsID	Gene	Function	CPP	Beta	P-value
1	rs6656401	CR1	Intron	0.119	-0.017	8.67E-15
1	rs7515905	CR1	Intron	0.206	-0.019	3.75E-15
1	rs1752684	CR1	Regulatory	0.125	-0.017	3.77E-15
1	rs679515	CR1	Intron	0.220	-0.018	3.60E-15
2	rs4663105	BIN1	Regulatory	0.631	0.050	1.26E-26
2	rs6733839	BIN1	Regulatory	0.796	0.053	1.24E-26
6	rs9270999	HLA-DRB1	Intron	0.181	0.001	8.04E-08
6	rs9273472	HLA-DRB1	Intron	0.110	0.074	1.63E-04
7	rs10808026	EPHA1	Intron	0.123	-0.020	1.36E-11
7	rs11762262	EPHA1	Intron	0.117	-0.011	2.21E-10
7	rs11763230	EPHA1	Intron	0.325	-0.020	1.86E-11
7	rs11771145	EPHA1	Intron	0.173	-0.021	8.69E-10
8	rs28834970	PTK2B	Intron	0.137	0.066	3.22E-09
8	rs2279590	CLU	Intron	0.166	0.021	4.47E-17
8	rs4236673	CLU	Intron	0.123	0.020	3.25E-17
8	rs11787077	CLU	Intron	0.247	0.022	2.94E-17
8	rs9331896	CLU	Intron	0.154	0.022	8.38E-17
8	rs2070926	CLU	Intron	0.278	0.023	2.69E-17
11	rs11039390	NUP160	Downstream	0.145	-0.004	2.31E-05
11	rs4939338	MS4A6E	Upstream	0.139	0.011	2.79E-12
11	rs7110631	PICALM	Intergenic	0.134	0.014	8.77E-15
11	rs10792832	RNU6-560P	Regulatory	0.633	0.027	7.89E-16
11	rs11218343	SORL1	Regulatory	0.643	-0.046	4.77E-11
14	rs10498633	SLC24A4	Intron	0.371	-0.059	1.55E-07
19	rs3752246	ABCA7	Missense	0.361	-0.027	4.27E-09
19	rs4147929	ABCA7	Regulatory	0.111	-0.030	1.77E-09
19	rs41289512	PVRL2	Regulatory	1.000	0.132	1.81E-167
19	rs6857	PVRL2	3' UTR	1.000	0.359	0
19	rs769449	APOE/TOMM40	Regulatory	1.000	0.292	0
19	rs56131196	APOC1	Regulatory	1.000	0.251	0
19	rs78959900	APOC1	Downstream	1.000	-0.096	8.22E-85
19	rs12459419	CD33	Missense	0.245	-0.027	6.66E-08

How to integrate such predicted functional annotations of non-coding variants with GWAS data?

- Variant-set genetic association test
  - Gene-set association test
  - Rare variant (RV) association test

STAAR: Variant-set genetic association test using functional annotation information



a, Input data of STAAR, including genotypes, phenotypes, covariates and (sparse) genetic relatedness matrix is prepared. **b**, All variants in the genome are annotated and the annotation principal components for different classes of variant function are calculated. PCA, principal component analysis. **c**, Two types of variant sets are defined: gene-centric analysis by grouping variants into functional genomic elements for each protein-coding gene; and genetic region analysis using agnostic sliding windows. **d**, The STAAR statistics for each variant set is calculated. aPC, annotation principal component. **e**, The STAAR-O *P* values for all variant sets defined in **c** are obtained and significant findings are reported. *Li, X. et. al. Nat. Genet. 2020.* https://www.nature.com/articles/s41588-020-0676-4

Application Study by STAAR:
Lipid traits in the TOPMed WGS data.

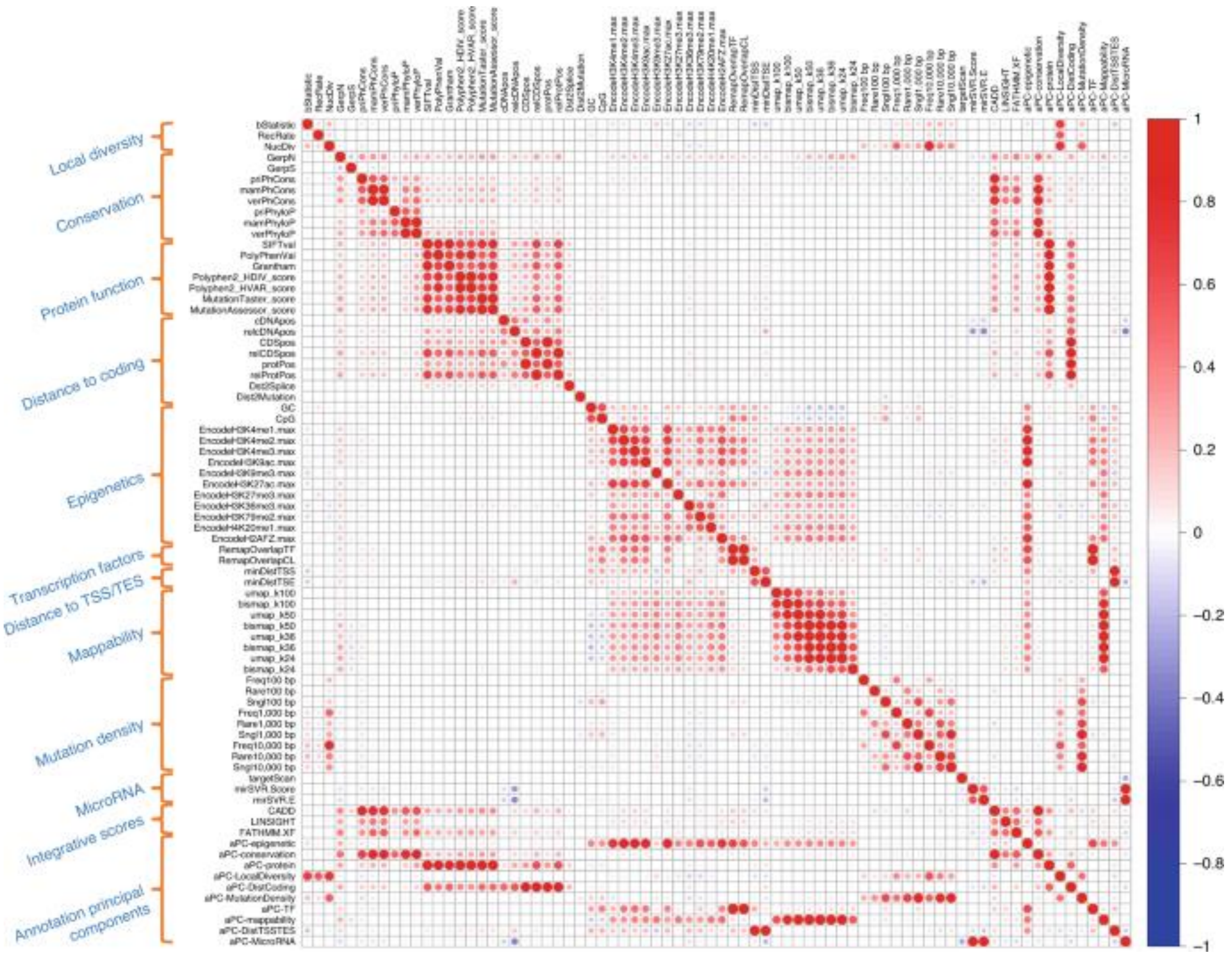
- Using TOPMed WGS data
- Study four quantitative lipid traits (LDL-C, HDL-C, TG and TC). LDL-C and TC were adjusted for the presence of medications.
- The discovery phase consisted of four study cohorts of TOPMed Freeze 3. The replication phase consisted of ten different study cohorts in TOPMed Freeze 5 that were not in Freeze 3.
  - 12,316 discovery samples, which had 155 million single-nucleotide variants (SNVs),
  - 17,822 replication samples, which had 188 million SNVs.

Application Study by STAAR:
Lipid traits in the TOPMed WGS data.

- STAAR-O method was used to perform
  - (1) gene-centric analysis using RV sets (MAF<1%, 143.2M) based on functional categories
  - (2) genetic region analysis using variant sets defined by 2-kb sliding windows with 1-kb skip length across the genome.
- Adjusted for age, age2 , sex, race/ethnicity, study and the first ten ancestral principal components
- Controlling for relatedness using linear mixed models
- Phenotypes were transformed by inverse rank normal transformation
- Annotations: 2 MAF weights, and 13 aggregated functional annotation scores:
  - 3 integrative scores (CADD, LINSIGHT, and FATHMM-XF)
  - 10 annotation principal components.

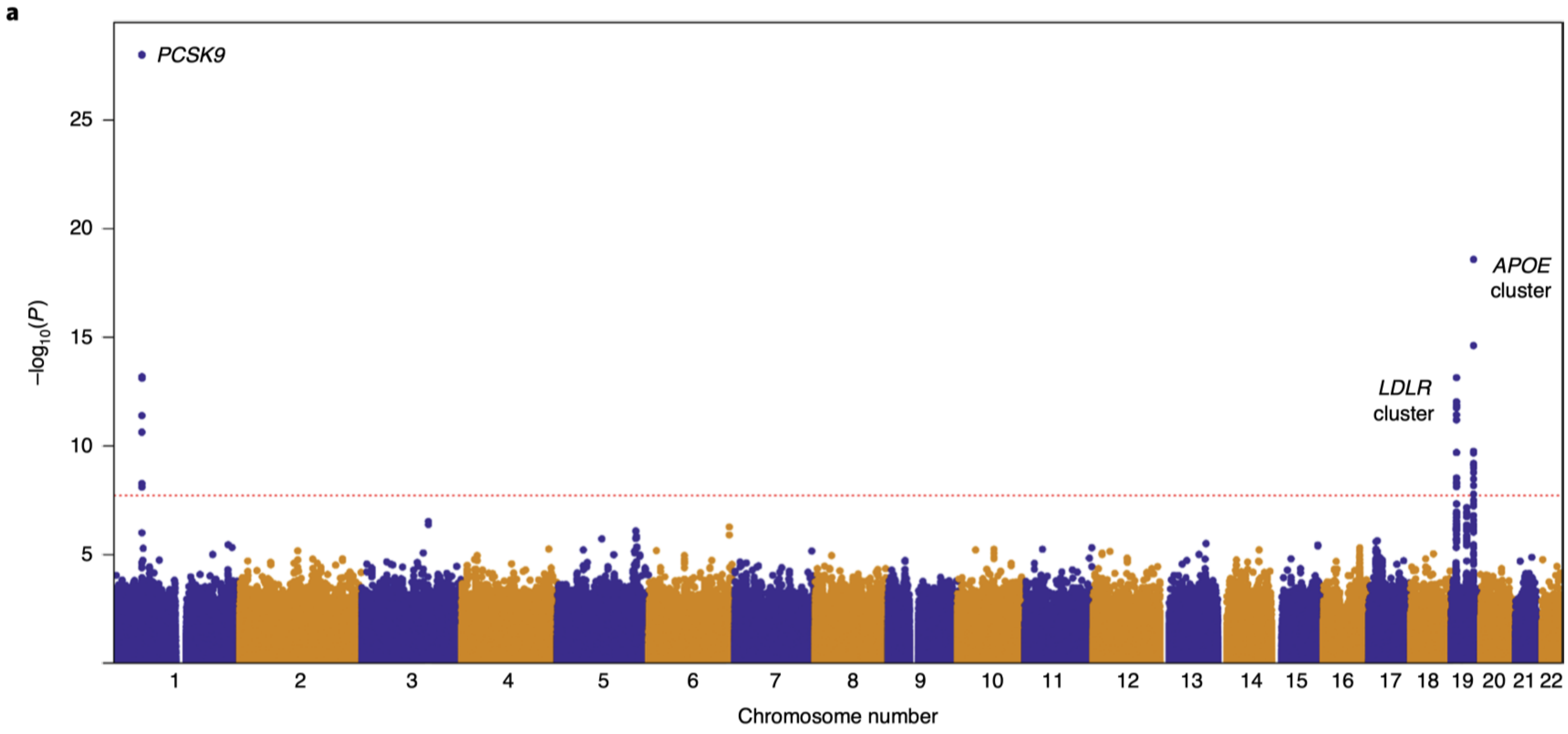
Correlation heatmap of functional annotation scores.

63 individual scores;
3 integrative scores;
10 annotation principal components.

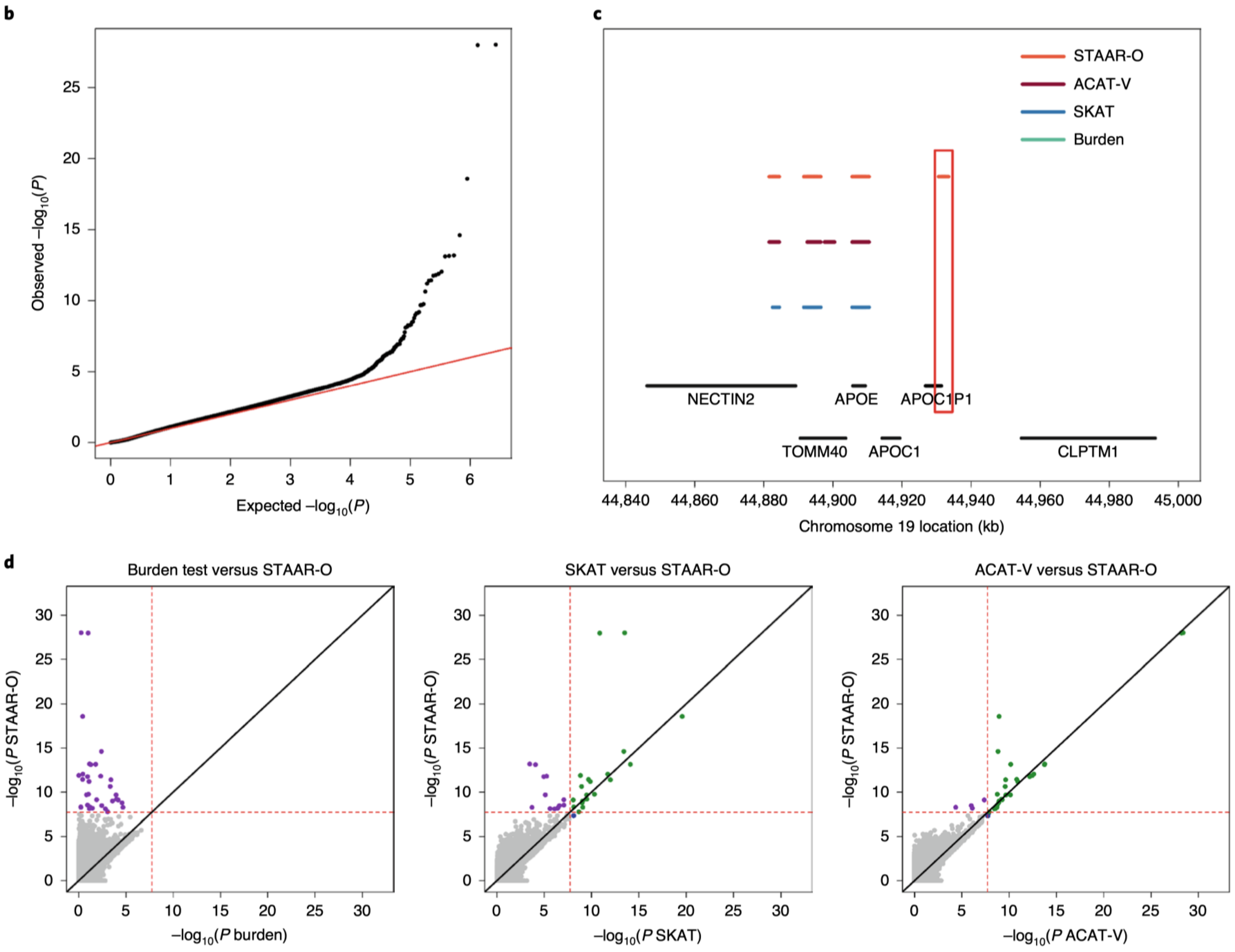


Genetic region (2-kb sliding window) unconditional analysis results of LDL-C in the discovery phase using the TOPMed cohort.

a, Manhattan plot showing the associations of 2.66 million 2-kb sliding windows for LDL-C versus −log10 (P) of StAAr-O. the horizontal line indicates a genome-wide P value threshold of 1.88 × 10E−8 (n = 12,316).

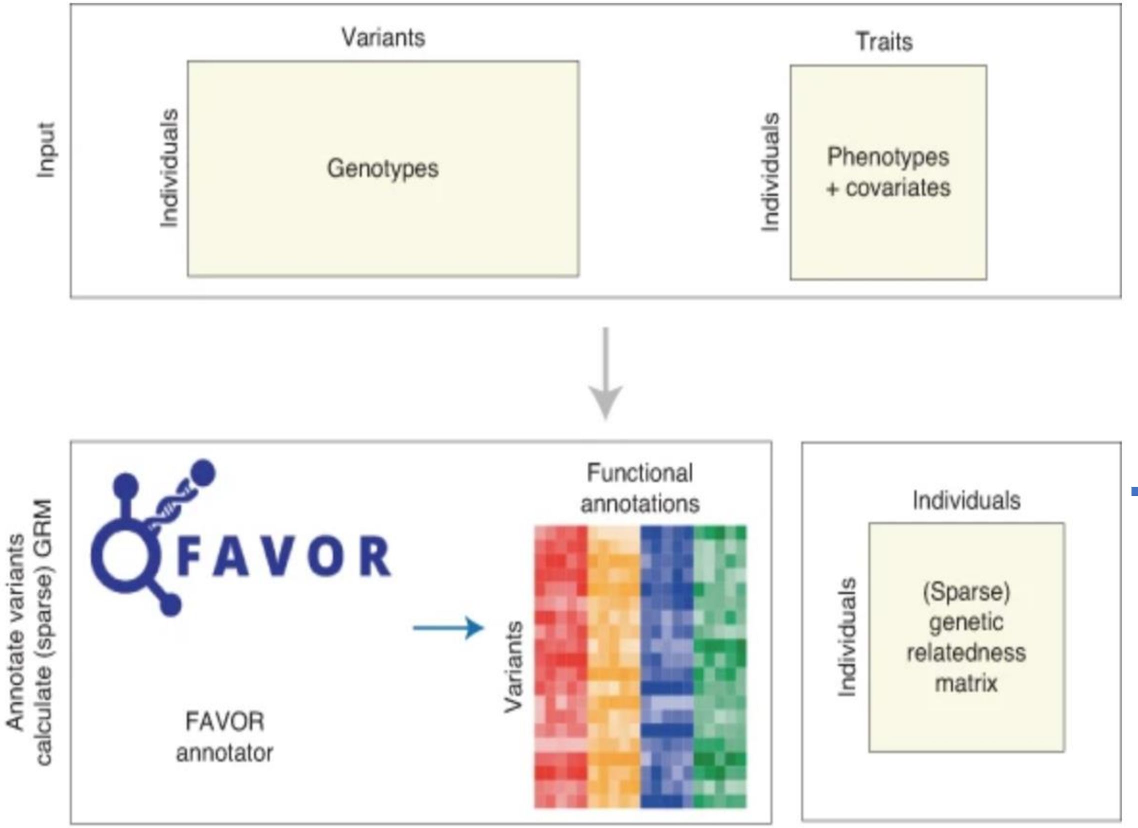


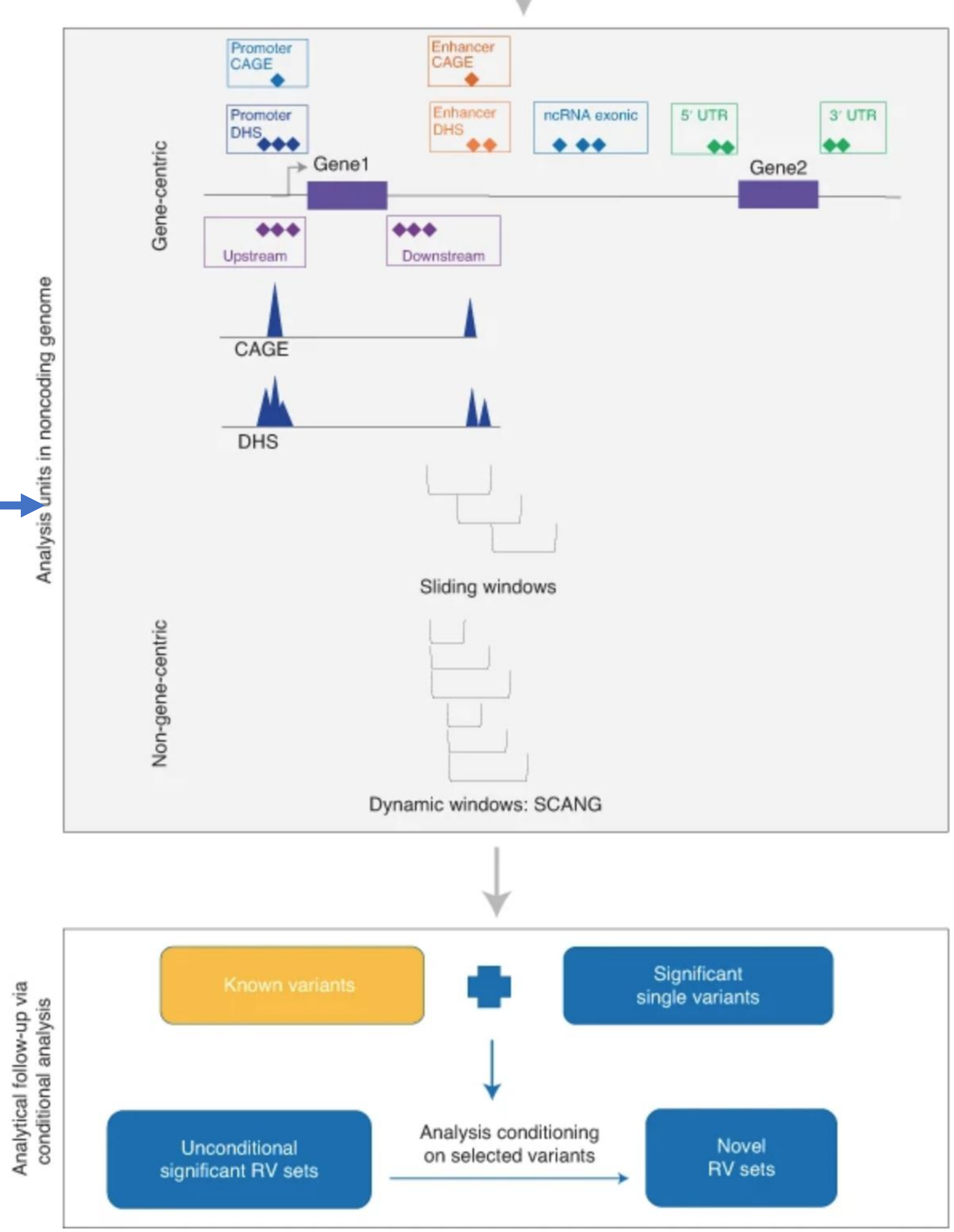
b, Quantile–quantile plot of 2-kb sliding window StAAr-O P values for LDL-C (n = 12,316).
c, Genetic landscape of the windows significantly associated with LDL-C that are located in the 150-kb region on chromosome 19. Four statistical tests were compared: burden; SKAT; ACAT-V; and STAAr-O. A dash indicates that the sliding window at this location was significant using the statistical test that the color of the dash represents (n = 12,316).
d, Scatterplot of P values for the 2-kb sliding windows comparing STAAR-O with the burden, SKAT and ACAT-V tests. each dot represents a sliding window with the x axis label being the −log10 (P) of the conventional test and the y axis label being the −log10 (P) of StAAR-O (n = 12,316).



STAARpipeline: an all-in-one rare-variant tool for biobank-scale whole-genome sequencing data

(https://www.nature.com/articles/s41592-022-01641-w)





- Provides single-variant analysis for common and low-frequency variants
- Gene-centric analysis for coding rare variants using multiple coding masks

Web Resources

- DEEPSEA Webtool
  - https://hb.flatironinstitute.org/deepsea/
- BFGWAS_QUANT
  - https://github.com/yanglab-emory/BFGWAS_QUANT
- STAAR
  - https://favor.genohub.org/
  - https://github.com/xihaoli/STAAR
  - https://github.com/xihaoli/STAARpipeline